The Impact of Tenure-Dependent Benefits on Politicians' Behavior and Government Stability

Click here for the latest version

Enrico Miglino*

September 26, 2024

Abstract

I combine newly digitized data on confidence votes, party switches and tax returns of Italian MPs between 2001-2022 to study how tenure-dependent benefits influence politicians' behavior and government stability. Using a difference-in-discontinuities design, I find that a tenure requirement for a parliamentary pension introduced in 2008 increased the probability of voting confidence in the government and switching to majority parties. It also incentivized legislative effort, as evidenced by a rise in approved bill proposals. In line with agency model predictions, the policy discouraged defiance of party directives from within, which could jeopardize reelection chances and pension eligibility in a second term. Tenure requirements increase government stability but also party control, with ambiguous effects on voters' welfare.

*Enrico Miglino: Bank of Italy (email: Enrico.Miglino@esterni.bancaditalia.it). I would like to thank the *Fondazione Rodolfo Debenedetti* and the *Fondazione Openpolis* for kindly granting access to their data. *The views expressed in this paper are the views of the author and do not involve the responsibility of the Bank of Italy.*

1 Introduction

> If a man wants to join our ranks, if he wishes to accept my modest program, to transform himself and become a progressive, how could I reject him?
>
> Agostino Depretis (Italian prime minister, 1882)

The parliamentary power to oppose governments is one of the checks and balances upon which modern democracies are based. But the policy uncertainty associated with divided governments and government crises discourage investment, hiring, lending, and ultimately hinders economic growth [Alesina et al., 1996; Bloom et al., 2007; Bloom, 2014; Baker et al., 2016; Bordo et al., 2016]. The critical decision to topple a government often lies in the hands of few members of Parliament (MPs), who can either be vocational and vote to maximize voters' welfare, or opportunistic and vote to extract tangible rents for themselves [Persson and Tabellini, 2000].

Parliamentary benefits may increase government stability in parliamentary democracies because they are an incentive for opportunistic MPs to hold their seat. But they can also affect the control that parties have over their MPs because they change reelection incentives. Even more so when benefits, such as pensions, depend on the number of years spent in Parliament. Tenure requirements for parliamentary pensions are common in modern democracies (Figure 1). For instance, U.S. Members of Congress are eligible for a congressional pension only after five years of service [Congressional Research Service, 2023]. Yet, their effects on politicians' behavior are surprisingly understudied.

This paper asks how tenure-dependent parliamentary benefits affect MPs' voting and party-switching decisions. To address this question, I study a change in monetary incentives occurred in Italy in 2008 (legislature XVI): the introduction of a minimum parliamentary tenure of 4.5 years required to obtain a parliamentary pension.

For this analysis, I have digitized the stenographic records of all confidence votes held in the Italian Parliament between legislatures XIV and XVIII (2001-2022), which contain the list of members of parliament who voted in favor, against or abstained. I then merged this data with individual-level information on all MPs, including their demographics, educational attainment, party affiliation, and the start and end dates of their parliamentary terms, enabling the calculation of their parliamentary tenure. The Parliament also provided data on legislative effort: the sponsors and date of each legislative bill proposal, their constitutional status, and whether they were approved. Finally, I digitized the first tax return submitted by each Deputy and Senator and kept in the parliamentary archives, to obtain a measure of their private income in the year prior to entering Parliament.

To study the effects of the policy, I cannot employ a regression discontinuity design because there is another monetary incentive, a parliamentary severance pay, that increases at the threshold of 4.5 years. I do not use a difference-in-differences design because newly-elected MPs may have a different trend in their voting attitude with respect to MPs with a much longer experience in Parliament. Therefore, I combine two sources of variation, before/after the beginning of Legislature XVI and just below/above 4.5 years parliamentary tenure, and implement a 'difference-in-discontinuities' design as in Grembi et al. [2016]. I take the difference between the pre-treatment and the post-treatment discontinuity at the tenure threshold in order to net out the effect of the severance pay increase. This strategy requires that the effect of confounding factors at the threshold does not vary over time.

I estimate that the introduction of the tenure requirement for a parliamentary pension increases the probability of expressing a vote of confidence in the government by 3 percentage points from an average of 70 percent of MPs voting confidence. To confirm the validity of the empirical strategy, I perform a series of diagnostic tests. First, the effect is not statistically different from zero in any of the Houses in the two legislatures (XIV and XV) before the introduction of the tenure requirement, but it becomes significantly positive since the first legislature in which the tenure requirement is in place (XVI). Secondly, the estimate is remarkably stable when using different bandwidths around the tenure threshold, from two months up to three years on each side. Thirdly, the Frandsen [2017] test for discrete running variables cannot reject the null hypothesis of absence of manipulation in the distribution of the tenure of MPs in confidence votes, and pre-determined variables do not show significant discontinuities. Finally, to assess the possibility that this result arises from random chance rather than from a causal relationship, I perform a set of estimations at placebo thresholds below and above 4 years and 6 months. All the placebo estimates are lower than the true-threshold coefficient for the confidence vote.

In a heterogeneity analysis, I find that the effect of the tenure requirement is stronger in legislatures in which MPs' private income and the chances of reelection are lower. Surprisingly, the policy significantly increases the votes of confidence by MPs elected in parties that support the government (*majority MPs*), but it decreases votes of confidence by MPs elected in opposition parties (*opposition MPs*) even if the latter effect is not significant in all specifications. As a consequence, the tenure requirement significantly reduces the probability that an MP votes against the directives of the electoral-affiliation party by 3.4 percentage points. These empirical results can be rationalized in a simple political-agency model in which policymakers are to some extent opportunistic.

First of all, if they were vocational and voted purely in their voters' interest, their voting behavior should not change when their private economic incentives change. Secondly, the model shows that the minimum tenure requirement for a parliamentary pension has two effects on the voting behavior of newly-elected MPs, which affect majority and opposition MPs differently. A 'pivotal' effect for which both majority and opposition newly-elected MPs have an incentive to vote confidence in case they are the pivotal voter so as to increase the probability of survival for the current government, complete their first legislature and secure a parliamentary pension. A 'party-discipline' effect for which both majority and opposition newly-elected MPs have an incentive to vote following the party directives so as to have a higher probability of being reelected (Table D) and reach the tenure requirement in their second term, in case the government loses the confidence vote and there are early elections. The party-discipline effect increases MPs' loyalty towards their own party, even if this is detrimental to voters' welfare.

Both effects go in the same direction for majority MPs, but they go in opposite directions for opposition MPs because the latter have a party directive to vote against the government. Then, the model predicts that the minimum tenure requirement will have an unambiguously positive impact on the probability to vote confidence for majority MPs, but it will have an ambiguous effect on opposition MPs, depending on which (if any) of the pivotal and party-discipline effect dominates. These predictions are confirmed by the heterogeneity analysis: the estimated effect is positive and highly significant for majority MPs, whereas it is negative or zero for opposition MPs (i.e. the party-discipline effect seems to dominate). The model shows that if the party-discipline effect dominates, the minimum tenure requirement is unambiguously distortionary: it induces majority MPs to vote in favor of the government and opposition MPs to vote against the government when it would be in the voters' interest to do otherwise.

A model extension shows that tenure-dependent benefits also act as a reelection incentive on legislative effort. They induce MPs to put more effort into signaling their legislative productivity in order to raise their nomination and reelection chances. In line with this prediction, the tenure requirement increases the daily probability of submitting a successful legislative bill proposal by 0.05 percentage points from an average of 0.03 percent. The reelection incentive also makes it more convenient for opposition MPs that want to support the government to switch party altogether, rather than defying party directives without changing affiliation. Empirically, I find that the tenure requirement more than doubles the likelihood of switching to a majority party, whereas switches to opposition parties are unaffected.

Even if parliamentary benefits have relatively small effects on the probability of voting confidence in a government, their impact on government stability and the macroeconomy of a country can be substantial. Back-of-the-envelope calculations show that one of the eight governments (Berlusconi IV) in legislatures XVI-XVIII would have resigned earlier, had the tenure requirement been absent. Interestingly, Italy experienced a severe sovereign debt crisis in the final months of this government, which the pension incentive helped to prolong.

This paper contributes to the literature on institutions and political stability [Diermeier and Merlo, 2000]. Empirical studies on political stability usually rely on strong identifying assumptions. Some notable exceptions are Gagliarducci and Paserman [2012], Acconcia and Ronza [2022] and Carozzi et al. [2022]. They employ RD designs to show that policymakers' gender and political fragmentation can significantly affect local government stability, whereas I show that monetary incentives based on parliamentary tenure can significantly affect government stability at a national level.

Second, the model and empirical results illustrate the presence of a trade-off between government stability and party control. According to Canen et al. [2020a,b], party discipline has been a growing driver of polarization in US congress since the 1970s, accounting for 65 percent of polarization in roll call voting in 2018. My paper shows that tenure-dependent benefits can enhance party control over the voting choices of their legislators, even when defying party directives would be beneficial for the general welfare.

Third, this work relates to the literature on politicians' incentives that tests the predictions of political-agency models [Besley and Case, 1995; Besley, 2004; Preece et al., 2004]. Starting from the seminal papers by Barro [1973] and Ferejohn [1986], these models assume that voters are 'principals' with imperfect information about the state of the world and that policymakers are opportunistic 'agents' working for them [Besley, 2007]. Recently, the focus has been on how relative salaries in the political and private sectors affect politicians' selection and behavior [Diermeier et al., 2005]. Most empirical analysis find positive effects of higher pay on politicians' quality, effort and performance [Ferraz and Finan, 2009; Kotakorpi and Poutvaara, 2011; Gagliarducci and Nannicini, 2013] with the exception of Fisman et al. [2015].[1] Rather than studying changes in relative wages, I analyze the effect of a perquisite attached to holding a parliamentary seat, the right for a parliamentary pension.[2] While the identification strategy of this paper does not allow to draw inference on the selection of legislators, it confirms the positive effects of monetary incentives on their legislative effort. More importantly, it provides new evidence that monetary incentives can affect politicians' behavior in high-stake decisions, such as party affiliation and confidence votes, with long-lasting consequences in terms of government stability.

The paper is organized as follows. Section 2 presents the institutional background of the pension reform. Section 3 describes the simple political-agency framework that guides the empirical analysis. Section 4 explains the empirical strategy and Section 5 illustrates the data. Section 6 presents the results, tests the validity of the empirical strategy and reports back-of-the-envelope calculations on the impact on government stability. Section 7 concludes.

[1] Theoretical results are inconclusive: if Caselli and Morelli [2004] find that higher salaries improve the quality of politicians assuming they have uni-dimensional ability, more complex models lead to ambiguous predictions due to free-riding effects [Messner and Polborn, 2004], the simultaneous presence of entry and retention effects [Mattozzi and Merlo, 2008] the presence of different types of politicians [Diermeier et al., 2005; Keane and Merlo, 2010].

[2] In this sense, the paper is close to the cross-sectional analyses by Hall and Van Houweling [1995] and Groseclose and Krehbiel [1994] which provide suggestive evidence of strategic retirement of US congressmen to take advantage of 'the golden parachute' provision in the Federal Election Campaign Act.

2 Institutional background

2.1 The Parliament and the votes of confidence

Italy is a parliamentary democracy with a bicameral structure. Until legislature XVIII, the Chamber of Deputies was composed by 630 elected Deputies and the Senate was composed by 315 elected Senators with the same legislative power. Following a constitutional referendum, in 2019 the Parliament approved a reform which reduced the number of Deputies to 400 Deputies and the number of Senators to 200 starting from the following legislature (XIX) [Dipartimento per le Riforme Istituzionali, 2022]. This reform reduced the probability that an MP will be reelected and this might affect the impact of the minimum tenure requirement for a pension in legislature XVIII.

All MPs are elected simultaneously during general political elections, except for Senators with a life tenure. These are former presidents of the Republic or citizens directly appointed by the president of the Republic 'for outstanding patriotic merits' (at most five). Regardless of the electoral-affiliation party and district, MPs have the legal duty to represent the interests of all Italian citizens. The parties form parliamentary groups whose heads determine the calendar of Parliament and the issues to be discussed during each parliamentary session. Yet, the parties have no formal control over the voting behaviour of the MPs while they are in Parliament [Merlo et al., 2010].

A majority in each House is required to pass a bill before it becomes a new law. Beyond the legislative power, the Parliament exercises control over the executive power of the Council of Ministers, primarily by means of a vote of confidence. There are several instances in which a vote of confidence can occur. First, before being officially in power, every Government must obtain the majority in each House through a vote of confidence. Second, each House can cast a vote of no confidence at any moment during a parliamentary term (legislature) as long as the no-confidence motion is signed by at least one tenth of the House members [Senato della Repubblica, 2022a]. Third, the Government may call a vote of confidence in order to compel the House to reconfirm its support in relation to a specific text being considered by the House and speed up the legislative procedure.

If the government loses a vote of confidence in any of the two Houses, the President of the Council of Ministers has to resign and the government falls [Camera dei Deputati, 2022]. No-confidence votes are not 'constructive' as in Spain and Germany: MPs do not propose an alternative candidate President of the Council who has to have a parliamentary majority and takes charge if the incumbent loses the vote. In this sense, the Italian confidence vote is similar to the confidence vote in the majority of parliamentary and semi-presidential democracies [Rubabshi-Shitrit and Hasson, 2022].[3]

[3] The constructive vote of no-confidence is currently present in seven countries: Germany, Spain, Hungary, Poland, Slovenia, Belgium and Israel. The vast majority adopts a regular vote of no-confidence as in Italy: Australia, Austria,

The constitutionally mandated duration of a legislature is five years. The President of the Republic can dissolve the Parliament before the natural end of a legislature and call for early elections if the Parliament is unable to form a stable majority in each House in support of a government.[4] This occurs generally after a loss in a vote of confidence (as for Romano Prodi's second government in January 2008) or after a win in a vote of confidence by an excessively narrow margin (as for Giuseppe Conte's second government in August 2019). In this case, parliamentary elections must be held within seventy days from the parliament dissolution (Italian constitution, article 61).[5]

Early elections have been relatively frequent in Italy. There have been eighteen legislatures between 1948 and 2022 and half of them ended before the natural term. The high degree of political instability in Italy resulted in high executive turnover: there have been 69 governments in the last 75 years, with an average duration of 1.1 years.

As we can see in Table 1, there were 223 votes of confidence/no confidence in the Chamber of Deputies and 204 votes of confidence/no confidence in the Senate, for a total of 427 in legislatures XIV-XVIII. Votes of confidence are quite frequent: the average distance between two votes of confidence is 38 days for the Senate and 35 for the Chamber, the maximum distance is 364 days for the Senate and 306 for the Chamber. Motions of no-confidence votes are included in this count but they are few: only five in the Senate and one in the Chamber in the analyzed period. Each legislature had between one and three different governments. Among the five legislatures under study, legislature XV and legislature XVIII ended before their natural end. The last vote of legislature XV was one year and nine months after its beginning, whereas the last vote of legislature XVIII was four years and four months after its beginning.

Bulgaria, Canada, Czech Republic, Croatia, Denmark, Estonia, Finland, Iceland, India, Ireland, Latvia, Lithuania, Netherlands, New Zealand, Norway, Portugal, Romania, Slovakia, Sweden and the United Kingdom [Rubabshi-Shitrit and Hasson, 2022].

[4]Article 88 of the constitution states that the President of the Republic cannot dissolve the Parliament "during the final six months of the presidential term, unless said period coincides in full or in part with the final six months of Parliament.". There were three presidents in the period of analysis: Ciampi (1999-2006), Napolitano (2006-2013; 2013-2015) and Mattarella (2015-2022; 2022-). The last semester of the first two presidents coincided with the final six months of Legislature XIV and XVI, respectively, so these presidents were allowed to dissolve the Parliament at any moment during their seven-year terms. Mattarella's last semester occurred between August 2021 and February 2022, which corresponds to the period between the fifth and the eleventh month of the third year of Legislature XVIII. The empirical results are robust to excluding Legislature XVIII from the analysis or to reducing the bandwidth to six months around the threshold: from the fourth to the fifth year of Legislature XVIII.

[5]The tenure period of an MP ends when the new MP is elected. When they cast a confidence vote, MPs do not know when the President of the Republic will set the new election date. According to a trembling hand equilibrium argument, MPs should consider the date of the last vote of confidence to assess their pension eligibility in order to be sure to obtain the parliamentary pension.

Table 1: Votes of confidence and majority margins in legislatures XIV-XVIII

Government	Legislature	Chamber votes	Senate votes	First vote	Last vote	Chamber majority	Senate majority
Berlusconi II	14	19	10	21/06/2001	28/12/2004	36	17
Berlusconi III	14	12	9	28/04/2005	09/02/2006	19	12
Prodi II	15	17	16	23/05/2006	24/01/2008	29	7
Berlusconi IV	16	32	23	15/05/2008	17/11/2011	20	15
Monti I	16	34	18	22/12/2011	21/12/2012	241	99
Letta I	17	10	7	30/04/2013	24/02/2014	138	75
Renzi I	17	31	42	26/03/2014	07/12/2016	63	2
Gentiloni I	17	14	20	14/12/2016	23/12/2017	53	11
Conte I	18	10	6	06/06/2018	05/08/2019	35	13
Conte II	18	21	24	10/09/2019	17/02/2021	28	11
Draghi I	18	22	29	25/02/2021	21/07/2022	216	64

Notes: This table shows the number of confidence votes in the Chamber and the Senate for each government in legislatures XIV-XVIII from 2001 to 2022. Each government is identified by the President of the Council and its ordinal number. *First vote* and *Last vote* indicate the date of the first and last confidence vote on each government. *Chamber majority* and *Senate majority* indicate the majority margin the government had in its first confidence vote in the Chamber and Senate, respectively.

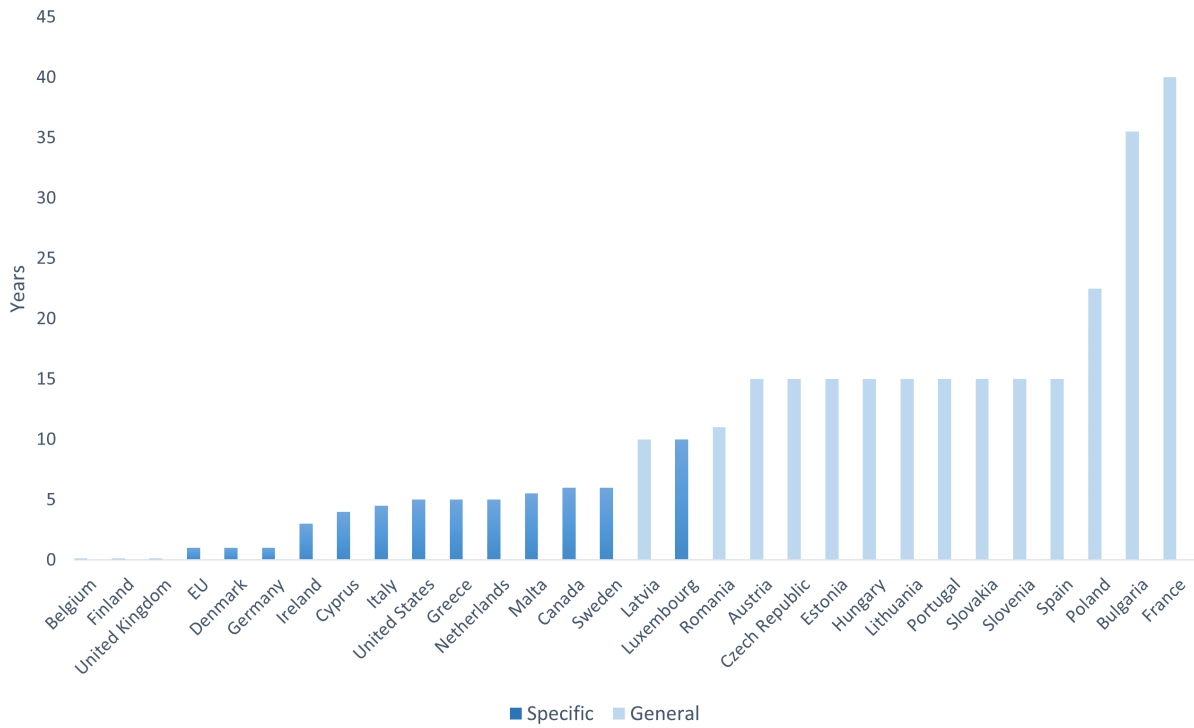
2.2 The parliamentary pension

A large group of countries have a minimum tenure requirement to obtain a parliamentary pension (Table C2). Some countries, such as Italy, have specific pension schemes dedicated to MPs only. Others, such as the United States, grant specific pensions to high-level civil servants, such as MPs and federal employees. In the remaining group of countries, such as France, MPs earn pension rights on the same terms as the rest of the labor market [Congressional Research Service, 2023; Office of the Superintendent of Financial Institutions Canada, 2019; European Parliament, 2021; House of Commons, 2019].

Most schemes covering MPs have short tenure requirements to account for the uncertain nature of political office. Figure 1 shows the minimum period of service as MP (in specific schemes) or the minimum period of employment (in general schemes) to qualify for a pension in 2009. When MPs participate in general schemes, their service as MP is counted as any other occupation and the minimum number of years of paid contributions to obtain a pension is generally higher, from 10 years in Latvia to 40 years in France. Specific schemes require shorter periods of service spent in Parliament, ranging from one month in the United Kingdom, 5 years in the United States, up to 10 years in Luxembourg.

In Italy, MPs have always been subject to a specific pension scheme and the parliamentary pension has always been cumulative with respect to any other pension received from other occupations [Rizzo, 2018].

Figure 1: Minimum tenure requirement for MPs' pension by country in 2009



Notes: This figure shows the minimum number of years that MPs are required to work to obtain a pension. In darker blue the countries in which MPs have a specific pension scheme and the minimum tenure requirement refers to the minimum number of years in Parliament. In lighter blue the countries in which MPs have a general pension scheme that applies to the entire population and years in parliament count as any other employment year. The sources of this data are the Congressional Research Service [2023], the Office of the Superintendent of Financial Institutions Canada [2019], the European Parliament [2021] and the House of Commons [2019].

Before 1997, Italian MPs received a parliamentary pension at the age of 60, after paying pension contributions for at least a five-year term in Parliament. The pension scheme was quite favorable: the pension amount corresponded to a share of the final salary (up to 85.5 percent), only in small part financed by the pension contributions. An MP that did not complete a five-year term could also obtain a pension by simply paying the missing monthly pension contributions. The facility to qualify for a parliamentary pension led to distortionary practices such as 'rotating resignations', which allowed multiple MPs to obtain a pension even after spending only few days in Parliament.

Since 1997, only MPs with more than 2.5 years of parliamentary tenure were allowed to pay the missing monthly pension contributions to obtain a pension [De Santis, 2020]. The minimum pension age was raised to 65 and decreased by one year for each year of parliamentary tenure after the first five years, down to a minimum of 60 years of age. The pension contribution was 8.6

percent of the gross parliamentary wage and the pension varied between 25 and 80 percent of the parliamentary monthly wage, increasing in tenure (from 5 to 15 years of tenure). In 2006 the gross parliamentary wage was €$12,434$ and therefore, the minimum pension was €$3,108$ with only 5 years of paid contributions.

In 2007, the right to pay the missing pension contributions to obtain a parliamentary pension was suppressed. The MPs elected for the first time since 2008 (from legislature XVI onwards) would receive a parliamentary pension only if they had more than 4.5 years of parliamentary tenure over their lifetime [Camera dei Deputati, 2007; Senato della Repubblica, 2022b].[6] The suppression of the pension redemption scheme for MPs followed a surge of an anti-politics sentiment in the Italian public opinion targeting MPs' privileges and has attracted considerable attention from the media [Fusani, 2007; Sesto, 2021]. If this reform was aimed at cutting the high costs of the Italian Parliament, several commentators warned about its distortionary incentive of voting in favor of a government to avoid the end of a parliamentary term before 4.5 years of tenure [De Santis, 2020; Osservatorio sui Conti Pubblici Italiani, 2021].

In addition, the monthly pension amount was changed, ranging between 20 and 60 percent of the final gross parliamentary monthly wage, increasing in tenure (from 5 to 15 years of tenure) [Camera dei Deputati, 2007]. The gross parliamentary monthly wage in legislature XVI was €$10,435$ and the pension contribution was 8.8 percent of the gross parliamentary wage. If the parliamentary term ended just before the MP could reach 4.5 years of tenure, the MP would lose a monthly pension of €$2,087$ from age 65 to the rest of his life as well as €$49,587$ in already paid pension contributions [Camera dei Deputati, 2013]. The minimum parliamentary pension was 57 percent higher than the average gross monthly pension in Italy in 2011 [ISTAT, 2013].

For MPs elected after January 1, 2012, the pension scheme changed from a final-salary pension scheme to a fully contributory pension scheme, in which the pension amount depends on the pension contributions. This resulted in a substantial pension cut from legislature XVI to legislatures XVII and XVIII. An MP reaching the minimum tenure threshold in legislature XVIII will earn only €970 when reaching 65 years of age [Daconto, 2022].[7]

Potential confounders. The parliamentary pension is not the only parliamentary benefit that changes at 4.5 years of tenure. At the end of a parliamentary term an MP receives a severance pay that increases by 80 percent of the monthly wage of the MP at 4.5 years of tenure [Camera

[6]Officially the minimum tenure is 5 years (i.e. one complete parliamentary term), but the eligibility threshold is actually 4 years, 6 months and 1 day as the tenure calculation approximates to the next semester [Osservatorio sui Conti Pubblici Italiani, 2021].

[7]For MPs who were already in parliament before 2012 a pro-rata system is applied: a final-salary pension scheme up to January 1, 2012 and a contributory pension scheme thereafter [Senato della Repubblica, 2022b]. After the retirement age parliamentary pension payments are temporarily suspended if the former MP is reelected in the national or European Parliament, in a Regional Council or is a member of the national government [Camera dei Deputati, 2013].

dei Deputati, 2001, 2022]. The severance pay has remain fixed at 80 percent of the monthly wage of the MP, multiplied by the number of years of parliamentary tenure, over the entire period of analysis.

The electoral system changed over the years of analysis. From 1993 to 2005, it was a system based on 75% single-member districts with a plurality rule and 25% proportional seats (Mattarella law). The second introduced in 2005 (Calderoli law) assigned all the seats proportionally, but the party or the coalition with a plurality of votes at the national or regional level would obtain a majority prize in the Chamber and Senate, respectively. Finally, the third introduced in 2017 (Rosato law) assigned two thirds of the seats in multi-member districts proportionally and the rest with a plurality rule in single-member districts. In all these years, voters could only vote for a party and could not express preferences on particular candidates within a party [Chiaramonte and d'Alimonte, 2018].

3 Conceptual framework

This section makes precise how the minimum tenure requirement for a parliamentary pension should affect MPs' voting behavior, using a simple political-agency model in the spirit of Besley [2007]. This conceptual framework yields several predictions that can be tested empirically.

Suppose that every politician is elected at the beginning of period 1, can be reelected in period 2, and retires in period 3.[8] Politicians elected in party j choose one binary action $c_{jt} \in \{0, 1\}$ in each of the first two periods, namely whether to vote confidence in the government or not. Assume that if a government loses the vote of confidence, the legislature ends before its natural term and there are early elections.[9]

Politicians are elected in majority parties M in favor of the government or in opposition parties m against the government: $i \in \{M, m\}$. Since the confidence vote of each MP is disclosed only after all MPs have voted, I assume that MPs do not observe the decisions of the other MPs when they vote. Let $\sigma(c_{j1})$ denote the probability that the government wins the confidence vote and the first legislature reaches its natural end, which is higher if the MP votes in favor: $\sigma(1) > \sigma(0)$.

Let $\pi_j(c_{jt})$ denote the probability that the MP is reelected and assume it is higher if they vote following party directives: $\pi_M(1) > \pi_M(0)$ and $\pi_m(0) > \pi_m(1)$. This assumption is based on the empirical evidence that first-term MPs who vote against their party directives are 10-25 percent less likely to be reelected (see Table C1 in Appendix Section D). Crucially, since 1993 voters

[8] 74 percent of the MPs in the sample remained in Parliament for no longer than two parliamentary terms, but the model can be extended to multiple periods without loss of generality.

[9] One could complicate the model and assume that losing the vote of confidence is associated with a positive (less than one) probability that the legislature ends prematurely. Under this assumption all the effects would be dampened, but the model would yield the same qualitative predictions.

cannot express a vote of preference for a particular candidate within a party in Italian parliamentary elections [Chiaramonte and d'Alimonte, 2018]. The candidates and the order of election are chosen by the respective parties. Therefore, voting according to voters' interest does not directly increase MPs' reelection probabilities in this model, but voting according to party directives does.

All politicians receive a per-period payoff $\mathcal{I}$ from holding a parliamentary seat (e.g. parliamentary wage, ego rents). If MPs are not reelected, they abandon the political career and earn a private wage w, which can be heterogeneous across politicians. When they retire, all MPs obtain a private pension proportional to their wage ξw. Assume politicians receive a parliamentary pension ρ only if they complete the first parliamentary term (i.e. the government wins the vote of confidence) or if they are elected for two parliamentary terms (even if incomplete).

When in Parliament, politicians observe the state of the world $s_t \in \{0, 1\}$, which is an indicator on the net welfare gain that the government is producing. If $s_t = 0$, it would be in the voters' interest for the Parliament to exercise its control power over the executive and topple the government. If $s_t = 1$, it would be in the voters' interest for the Parliament to guarantee government stability and vote confidence in the government. Voters receive a payoff Ω only if the politician they have elected votes according to the state of the world: i.e. if $c_{jt} = s_t$. If $c_{jt} \neq s_t$, voters receive 0. Politicians care about their voters' welfare up to a certain extent, measured by a parameter α. I abstract away from considerations on the relative wage and assume that politicians always prefer the political career to the private sector career: $\alpha\Omega + \mathcal{I} \geq w$.

Politicians have a time-additive intertemporal utility function, discount future by a factor $\beta < 1$ and the within-period utility function $u(\cdot)$ is increasing and concave. Note that, if reelected, the second-period utility of an MP is independent of her majority-opposition status and politicians always act in the voters' interest since it is their last term: second-period utility for a reelected MP is always $u(\mathcal{I} + \alpha\Omega)$.

MPs elected in a party $j \in \{m, M\}$ in period 1 choose their confidence vote c_{j1} under state of the world s_1 to maximize expected lifetime utility:

$$\begin{aligned}\max_{c_{j1}} V_j(c_{j1}) =& u(\mathcal{I} + \alpha[s_1 c_{j1} + (1 - s_1)(1 - c_{j1})]\Omega) + \beta\{\pi_j(c_{j1})u(\mathcal{I} + \alpha\Omega) + [1 - \pi_j(c_{j1})]u(w)\} \\ &+ \beta^2\{\gamma_j(c_{j1})u(\rho + \xi w) + [1 - \gamma_j(c_{j1})]u(\xi w)\}\end{aligned} \tag{1}$$

where $\gamma_j(c_{j1})$ is the probability of reaching the tenure requirement to obtain a parliamentary pension. If there is no tenure requirement (i.e. pre-reform) or if the vote c_1 occurs after MP j has reached the tenure requirement, we have $\gamma_j^{Pre}(c_{j1}) = 1$. The MP has a guaranteed parliamentary pension at the moment of the confidence vote.

If there is a tenure requirement (i.e. post-reform) and the vote c_1 occurs before the MP reaches the tenure requirement, we have $\gamma_j^{Post}(c_{j1}) = \sigma(c_{j1}) + [1 - \sigma(c_{j1})]\pi_j(c_{j1})$. After the reform, the probability the MPs will obtain a parliamentary pension is equal to the probability that the first legislature does not end prematurely plus the probability of being reelected in case the government falls.

The change in the utility increase for voting confidence in the government due to the introduction of the minimum tenure requirement is:

$$\begin{aligned}\Delta V_j^{Post} - \Delta V_j^{Pre} =& \beta^2[u(\rho + \xi w) - u(\xi w)]\Delta\gamma_j^{Post} = \\ & \beta^2[u(\rho + \xi w) - u(\xi w)]\{\Delta\sigma[1 - \pi_j(0)] + \Delta\pi_j[1 - \sigma(1)]\}\end{aligned} \tag{2}$$

where $\Delta V_j^s \equiv V_j^s(1) - V_j^s(0)$ is the increase in expected utility for voting confidence relative to voting against before ($s = Pre$) or after ($s = Post$) the reform. $\Delta\gamma_j^{Post} \equiv \gamma_j^{Post}(1) - \gamma_j^{Post}(0)$, is the increase in probability of reaching the tenure requirement for voting confidence relative to voting against after the reform. $\Delta\sigma \equiv \sigma(1) - \sigma(0)$ and $\Delta\pi_j \equiv \pi_j(1) - \pi_j(0)$ are the increase in probability of government survival and of reelection for voting confidence relative to voting against.[10]

The introduction of the minimum tenure requirement generates an overall incentive on MPs' confidence votes that is determined by the change in probability of obtaining a parliamentary pension $\Delta\gamma_j^{Post}$ multiplied by the discounted utility gain associated with the parliamentary pension $\beta^2[u(\rho + \xi w) - u(\xi w)]$. This overall incentive can be decomposed into two effects on politicians' behavior.

The first term in curly brackets ($\Delta\sigma[1 - \pi_j(0)]$) is a *pivotal effect*: the incentive to vote confidence in order to increase the probability of government survival, complete the legislature and secure a parliamentary pension in the first term. This effect becomes smaller the higher the probability of being reelected.

The second term in curly brackets ($\Delta\pi_j[1 - \sigma(1)]$) is a *party-discipline effect*: the incentive to vote following party directives so as to increase the probability of being reelected and reach the tenure requirement in the second legislature, in case the government falls and there are early elections. This effect becomes smaller the higher the probability of government survival.

Both pivotal and party-discipline effects decrease when the government has a larger majority margin, as each MP is less likely to be pivotal and the probability of a government fall is lower. The effects decrease in private wage w because of diminishing marginal utility and increase in age because of lower discounting.

The pivotal effect and the party-discipline effects are both positive for majority MPs ($\Delta\pi_M > 0$ and $\Delta\sigma > 0$). The tenure requirement unambiguously incentivizes MPs' to vote confidence in the

[10] This simple model abstracts away from general equilibrium considerations. Formally, I assume that $\sigma^{Pre}(c_{j1}) = \sigma^{Post}(c_{j1}) = \sigma(c_{j1})$ and $\pi^{Pre}(c_{j1}) = \pi^{Post}(c_{j1}) = \pi(c_{j1})$.

government before the threshold is reached, because voting confidence increases the probability of government survival and also their reelection probability.

The pivotal effect and the party-discipline effects have opposite signs for opposition MPs ($\Delta\pi_M > 0$ and $\Delta\sigma < 0$). The minimum tenure requirement for the parliamentary pension has an ambiguous effect on the behavior of newly-elected opposition MPs for voting confidence. If the incentive to increase the chances of government survival to obtain the pension in the first term is lower than the fear of losing the possibility of being reelected and obtain the pension later in case of government defeat ($\Delta\sigma[1 - \pi_j(0)] \leq \Delta\pi_j[1 - \sigma(1)]$), then the minimum tenure requirement incentives newly-elected opposition MPs to vote no confidence.

Majority MPs always vote confidence in the first period following the party directives if they are uninterested in voters' welfare ($\alpha = 0$) and always vote according to government performance if they are eminently interested in voters' welfare ($\alpha \to \infty$). If the party-discipline effect dominates, opposition MPs always vote no confidence in the first period following the party directives if they are uninterested in voters' welfare ($\alpha = 0$) and always vote according to government performance if they are eminently interested in voters' welfare ($\alpha \to \infty$). In these extreme cases (fully 'opportunistic' or fully 'vocational' politicians), the minimum pension requirement has no effect on their voting behavior (Appendix Section A).

We can summarize these theoretical results in the following empirical predictions.

Prediction 1: *The minimum tenure requirement increases votes of confidence by newly-elected majority MPs if they are sufficiently, but not fully opportunistic. This effect increases in the probability of being reelected and age, whereas it decreases in private wage and in the government majority margin.*

Prediction 2: *The minimum tenure requirement has an ambiguous effect on the votes of confidence by newly-elected opposition MPs. The effect is negative if the party-discipline effect dominates the pivotal effect. Opposition MPs are less likely to vote confidence if they are more likely to be reelected. Larger government majority margins and private wage weaken the effects, whereas age strengthens them.*

Normative implications. If voters have no preference for government stability, Appendix Section A shows that the minimum tenure requirement is a distortionary incentive for majority MPs because it induces them to vote in favor of the government, when it would be in the voters' interest to vote against. If the party-discipline effect dominates, the minimum tenure requirement is also a distortionary incentive for opposition MPs because it induces them to vote against the government, when it would be in the voters' interest to vote in favor.

Let $Pr(c_{j1}|s_1)$ denote the probability an MP elected in party $j \in \{m, M\}$ votes c_{j1} conditional on government performance being s_1 and let $Pr(s_1)$ denote the probability that government performance is s_1. The expected social surplus generated by MP elected in party $j \in \{m, M\}$ is:

$$E(W_j) = \Omega\{\beta\pi_j(0) + Pr(s_1 = 0) + Pr(s_1 = 1)Pr(c_{j1} = 1|s_1 = 1)[1 + \beta\Delta\pi_j] \\ - Pr(s_1 = 0)Pr(c_{j1} = 1|s_1 = 0)[1 - \beta\Delta\pi_j]\} \tag{3}$$

Intuitively, the expected social surplus increases in the probability of voting confidence in a good government and decreases in the probability of voting confidence in a bad government.

Appendix Section A shows that majority MPs always vote confidence when the government performs well, independently of the tenure requirement: $Pr(c_{M1} = 1|s_1 = 1) = 1$. Since the tenure requirement induces majority MPs to vote confidence: it increases $Pr(c_{M1} = 1|s_1 = 0)$. Therefore, the tenure requirement lowers the expected social surplus generated by majority MPs.

Without the tenure requirement, opposition MPs always vote no confidence when the government is not doing well: $Pr(c_{m1} = 1|s_1 = 0) = 0$. The tenure requirement induces opposition MPs to vote no confidence when the party-discipline effect dominates: $Pr(c_{m1} = 1|s_1 = 1)$ decreases Therefore, the tenure requirement also lowers the expected social surplus produced by opposition MPs if the party-discipline effect dominates. According to the definition by Besley [2007], if the party-discipline effect dominates, the minimum tenure requirement is a political failure as it produces a negative ex-ante social surplus. The model shows an example in which political survival considerations can be a source of real inefficiencies, as summarized by Besley and Coate [1998].

This normative analysis On the other hand, when government stability is positively valued by voters (Appendix Section B), the minimum tenure requirement decreases ex-ante social surplus only if voters' preference for government stability is sufficiently low.

4 Empirical strategy

4.1 Identification

I closely follow the identification strategy by Grembi et al. [2016]. Given the institutional background described above, there are three different treatments: the severance pay that increases at the threshold, the pension eligibility that changes at the threshold only after 2008, and the changes in the pension amount and electoral laws which applied uniformly on both sides of the threshold. Define: D_{it} as the first treatment for MP i at time t, equal to one if the severance pay is lower and zero otherwise; R_{it} as the second treatment, equal to one if the MP does not obtain the parliamentary pension and zero otherwise; A_t as the third treatment, equal to 1 if the pension amount and the electoral law are the ones that apply after 2008 and 0 if they are the ones that apply before 2008. The additional confounding treatment A_t differentiates this setting from the one in Grembi et al. [2016].

MPs with parliamentary tenure Z_{it} at or below the threshold Z_c (4.5 years) have a lower severance pay, while the pension eligibility requirement is introduced at time t_0 (year 2008) for MPs with tenure at or below the same threshold. Finally, the monthly pension amount decreased from 25 to 20 percent of the final gross parliamentary monthly wage at time t_0 on both sides of the threshold. The assignment mechanism for the three treatments can be formalized as below:

$$D_{it} = \begin{cases} 1 & \text{if } Z_{it} \leq Z_c \\ 0 & \text{otherwise,} \end{cases}$$

$$R_{it} = \begin{cases} 1 & \text{if } Z_{it} \leq Z_c \text{ and } t \geq t_0 \\ 0 & \text{otherwise,} \end{cases}$$

$$A_t = \begin{cases} 1 & \text{if } t \geq t_0 \\ 0 & \text{otherwise.} \end{cases}$$

Define $Y_{it}(d, r, a)$ as the potential policy outcomes if $D_{it} = d$, $R_{it} = r$, and $A_t = a$ with $d, r, a \in \{0, 1\}$. The observed outcome is equal to

$$\begin{aligned} Y_{it} =& D_{it}R_{it}A_tY_{it}(1,1,1) + D_{it}R_{it}(1-A_t)Y_{it}(1,1,0) + D_{it}(1-R_{it})A_tY_{it}(1,0,1) + \\ & D_{it}(1-R_{it})(1-A_t)Y_{it}(1,0,0) + (1-D_{it})R_{it}A_tY_{it}(0,1,1) + (1-D_{it})R_{it}(1-A_t)Y_{it}(0,1,0) \\ & + (1-D_{it})(1-R_{it})A_tY_{it}(0,0,1) + (1-D_{it})(1-R_{it})(1-A_t)Y_{it}(0,0,0) \end{aligned}$$

The objective is to identify the causal effect of R_{it} on Y_{it}. For ease of notation, let $W^- \equiv \lim_{z \to Z_c^-} E[W_{it}|Z_{it} = z, t \geq t_0]$ and $W^+ \equiv \lim_{z \to Z_c^+} E[W_{it}|Z_{it} = z, t \geq t_0]$ with $W \in \{Y, Y(1,1,1), Y(1,1,0), Y(1,0,1), Y(0,1,1), Y(1,0,0), Y(0,1,0), Y(0,0,1), Y(0,0,0)\}$.

In this setting standard continuity conditions are not sufficient for identification because of the confounding treatment D_{it}. Even assuming that all potential outcomes $E[Y_{it}(d, r, a)|Z_{it} = z, t \geq t_0]$ with $w, r, a \in \{0, 1\}$ are continuous in z at Z_c, we have that the cross-sectional RD estimator after t_0, $\hat{\tau}_{RD} \equiv Y^- - Y^+$, does not identify an average treatment effect of R_{it} at the threshold:

$$\begin{aligned} \hat{\tau}_{RD} &\equiv Y^- - Y^+ = Y(1,1,1)^- - Y(0,0,1)^+ \\ &= [Y(1,1,1)^- - Y(1,0,1)^-] - [Y(1,0,1)^+ - Y(0,0,1)^+] \\ &= E[Y(1,1,1)_{it} - Y(1,0,1)_{it}|Z_{it} = Z_c, t \geq t_0] - [Y(1,0,1)_{it} - Y(0,0,1)_{it}|Z_{it} = Z_c, t \geq t_0] \end{aligned}$$

where the first term in the right-hand-side captures one of the potential causal effects of interest (namely, the average treatment effect of establishing a minimum tenure requirement for a parliamentary pension for MPs in 2008 with a severance pay equal to 320 percent of the final wage and a monthly parliamentary pension that is 20 percent of the final wage) and the second term captures the 'bias' (namely, the average treatment effect of increasing the severance pay from 320 to 400 percent of the final wage for MPs with a monthly parliamentary pension that is 20 percent of the final wage). Accordingly, the cross-sectional RD estimate is biased because the effects of the two treatments D and R cannot be disentangled from each other.

Information on the pre-treatment period ($t < t_0$) allows to remove the selection bias under local assumptions. Similarly to the post-treatment period, for the pre-treatment period let $\tilde{W}^- \equiv \lim_{z \to Z_c^-} E[\tilde{W}_{it}|Z_{it} = z, t < t_0]$ and $\tilde{W}^+ \equiv \lim_{z \to Z_c^+} E[\tilde{W}_{it}|Z_{it} = z, t < t_0]$ with $\tilde{W} \in \{Y, Y(1,1,1), Y(1,1,0), Y(1,0,1), Y(0,1,1), Y(1,0,0), Y(0,1,0), Y(0,0,1), Y(0,0,0)\}$.

To identify the causal effect of eliminating the parliamentary pension under a certain parliamentary tenure, I exploit both the discontinuous variation at Z_c and the time variation after t_0 using a 'difference-in-discontinuities' estimator $\hat{\tau}_{DD}$:

$$\hat{\tau}_{DD} \equiv (Y^- - Y^+) - (\tilde{Y}^- - \tilde{Y}^+) \tag{4}$$

The identification assumptions for the 'difference-in-discontinuities' design are:

Assumption 1 *All potential outcomes $E[Y_{it}(d,r,a)|Z_{it} = z, t \geq t_0]$ and $E[Y_{it}(d,r,a)|Z_{it} = z, t < t_0]$ with $d, r, a \in \{0,1\}$ are continuous in z at Z_c*

Assumption 2 *The effect of the confounding policy D_{it} at Z_c in the case of no treatment ($R_{it} = 0$) is constant over time and does not depend on A_t: $Y(1,0,1) - Y(0,0,1) = \tilde{Y}(1,0,0) - \tilde{Y}(0,0,0)$.*

Assumption 2 requires that the effect of the severance pay discontinuity D_{it} at the threshold Z_c does not vary with time nor with the pension amount or the electoral laws. This is similar to the standard identifying assumption for diff-in-diff: it requires observations just below and just above Z_c to be on a local parallel trend in the absence of the policy of interest R_{it}. To indirectly test for this assumption, I estimate the pattern of the discontinuities in Y_{it} before t_0 and show that observations just below and just above Z_c were not on differential trends before the adoption of the minimum tenure requirement.

Under these two assumptions, the diff-in-disc estimator identifies the local causal effect of eliminating the parliamentary pension in a neighborhood of the tenure threshold ($Z_{it} = Z_c$), for MPs with a severance pay that is 320 percent of the final wage ($D_{it} = 1$) and with a monthly parliamentary pension equal to 20 percent of the final wage ($A_t = 1$):

$$
\begin{aligned}
\hat{\tau}_{DD} &\equiv (Y^- - Y^+) - (\tilde{Y}^- - \tilde{Y}^+) \\
&= [Y(1,1,1)^- - Y(0,0,1)^+] - [\tilde{Y}(1,0,0)^- - \tilde{Y}(0,0,0)^+] \\
&= [Y(1,1,1) - Y(0,0,1)] - [\tilde{Y}(1,0,0) - \tilde{Y}(0,0,0)] \\
&= [Y(1,1,1) - Y(1,0,1)] \\
&= E[Y(1,1,1)_{it} - Y(1,0,1)_{it} | Z_{it} = Z_c]
\end{aligned}
$$

Proposition 1 *Under Assumption 1 and Assumption 2, the diff-in-disc estimator $\hat{\tau_{DD}}$ identifies the average treatment effect at Z_c: $E[Y_{it}(1,1,1) - Y_{it}(1,0,1)|Z_{it} = Z_c]$.*

This result allows to identify a causal effect of the treatment of interest under plausible conditions. Yet, the estimand only refers to MPs with a severance pay equal to 320 percent of the final parliamentary wage. To identify a more general estimand with the diff-in-disc estimator, we can introduce an additional assumption.

Assumption 3 *The effect of the treatment R_{it} at Z_c does not depend on the confounding policy D_{it}: $Y(1,1,1) - Y(1,0,1) = Y(0,1,1) - Y(0,0,1) \equiv E[Y_{it}(1) - Y_{it}(0)|Z = Z_c, A_t = 1]$*

Assumption 3 states that there must be no interaction between the effect of the severance pay policy and the effect of the parliamentary pension policy. In my institutional setting, this assumption would be violated if MPs just below and just above Z_c, who receives a different severance pay, reacted to the minimum tenure requirement for the pension in different ways. In Section E I indirectly test this assumption by showing that the confounding policy (the severance pay) has no meaningful impact on the voting behavior of MPs. There is no significant discontinuity in the MPs' voting behavior at the 4.5 year threshold before the introduction of the minimum tenure requirement for a parliamentary pension.

4.2 Estimation

Let t_0 be the time in which the reform came into force (May 14, 2008). First, I can restrict the panel to the confidence votes occurred after (before) the reform $t \geq t_0$ ($t < t_0$) and implement a local linear regression:

$$
Y_{ipgt} = \delta_0 + \delta_1 \tilde{Z}_{it} + D_{it}(\pi_0 + \pi_1 \tilde{Z}_{it}) + \eta_{pg} + \phi_i + \varepsilon_{ipgt} \tag{5}
$$

where D_{it}, is an indicator for tenure of MP i at time t less than or equal to 4.5 years capturing treatment status, $\tilde{Z}_{it} = Z_{it} - Z_c$ is the parliamentary tenure of the MP centered at the threshold,

η_{pg} are party-by-government fixed effects, which control for the average support that the electoral-affiliation party p gives to the government g, and ϕ_i are MP fixed effects. The coefficient π_0 is the RD estimator and identifies the local treatment effect of risking to lose the parliamentary pension. These separate regressions allow to test whether the severance pay had a significant impact on the MPs' voting behavior before t_0 or whether the change in voting behavior is driven by the introduction of the tenure requirement for a parliamentary pension after t_0.

To disentangle the two effects, I implement a difference-in-discontinuities following Grembi et al. [2016]. The method consists in fitting linear regression functions to the votes distributed within a tenure window h on either side of the tenure threshold Z_c, both before and after t_0. Formally, we restrict the sample to votes in the tenure interval $Z_{it} \in [Z_c - h, Z_c + h]$ and estimate the model:

$$Y_{ipgt} = \zeta_0 + \zeta_1 \tilde{Z}_{it} + D_{it}(\theta_0 + \theta_1 \tilde{Z}_{it}) + Post_t[\alpha_0 + \alpha_1 \tilde{Z}_{it} + D_{it}(\beta_0 + \beta_1 \tilde{Z}_{it})] + \eta_{pg} + \phi_i + \varepsilon_{ipgt} \quad (6)$$

where $Post_t$ is an indicator for being elected in the post-treatment period $t \geq t_0$.[11] The coefficient β_0 is the diff-in-disc estimator and identifies the treatment effect of risking to lose the parliamentary pension, as the treatment is $D_i \cdot Post_t$. In all the tables, I show the results without fixed effects, with party-by-government fixed effects only, and with all fixed effects. My preferred specification is the latter, because it does not depend on the variation in the composition of the Parliament: it can control for all time-invariant unobserved heterogeneity across MPs, taking care of any potential unobserved imbalance. This specification captures the change in voting behavior within each MP voting for the same government at the tenure threshold, which is the aim of the empirical analysis.[12]

When you add party-by-government fixed effects you exploit the variation between members of the same party voting for the same government with different tenure. The addition of party-by-government fixed effects accounts for changes in government during the parliamentary tenure within the bandwidth around the threshold. For instance from Renzi to Gentiloni in December 2016.

When you add individual fixed effects you exploit the variation in the voting of the same MP voting for the same government before and after the threshold is reached. The addition of individ-

[11] I cannot use the 2.5-year threshold because of data limitations: my dataset does not contain votes of confidence before 1997, when this threshold was introduced. Therefore, at this threshold I cannot disentangle the minimum tenure requirement effect from the severance pay effect.

[12] To identify and estimate the coefficient of interest the specification with all fixed effects exploits the variation of votes within MPs that participated in at least one confidence vote in each side of the threshold within the bandwidth. These constitute 82 percent of the observations in the regression sample. If I keep only these MPs in the sample, the estimates are virtually unchanged.

ual fixed effects accounts for changes in the composition of MPs during the parliamentary tenure within the bandwidth around the threshold, due to to new elections or by-elections.

To account for within-MP autocorrelation, I cluster standard errors at MP level. As recommended by Kolesár and Rothe [2018], I do not cluster standard errors by the running variable as this results in confidence intervals with poor coverage property. Each regression uses uniform kernels. Because there is no clear way to determine the optimal bandwidth in the case of a discrete running variable, I present the robustness of the results to multiple bandwidths from two months up to three years on each side of the cutoff [Iizuka et al., 2021]. Following Lee and Lemieux [2010]; Gelman and Imbens [2019], my preferred estimation method is local linear regression, with different linear terms on the running variable estimated at either side of the threshold, but the main results are qualitatively unchanged when using local quadratic regressions.[13]

5 Data

Each of the two Houses of Parliament provide information on demographics and other characteristics of all members elected in each legislature since the inception of the Italian Republic in 1948. This data includes the name, surname, gender, date and town of birth, level of education and previous job, start date and end date of each parliamentary term, the parliamentary group (party) to which each MP is affiliated and the start-date and end-date of the party affiliation during a legislature. A party switch is defined as the within-legislature change in affiliation from one parliamentary group to another, including changes due to parliamentary-group merges, splits and dissolutions into the mixed group. I exclude switches due to cosmetic changes in party names and switches across legislatures. The parliamentary tenure of an MP corresponds to the total days spent in any of the two Houses. Therefore, I have merged the Deputies' and Senators' datasets using the name and surname of the MPs as the linking variable in order to construct an accurate measure of the parliamentary tenure of each MP, by summing the duration of all their parliamentary terms in both Houses. The Parliament also provides measures of legislative output: for each legislative bill proposal, the date of presentation, the author (*primo firmatario*), whether it was an ordinary or constitutional bill, and whether it was approved.

The two Houses kept the stenographic record of each parliamentary session in all the legislatures from XIV to XVIII. I have systemically searched for each vote of confidence and motion of no confidence occurred during these legislatures. The votes of all MPs are revealed at the end of a confidence vote. Therefore, the stenographic records contain a list of all the MPs who voted in

[13] Gelman and Imbens [2019] show that controlling for high-order polynomials in regression discontinuity analysis leads to noisy estimates, sensitivity to the degree of the polynomial, and poor coverage of confidence intervals. They recommend instead to use estimators based on local linear or quadratic polynomials.

favor, against or abstained for each vote of confidence. The non-listed MPs decided not to vote. I have digitized these votes from the original stenographic records and created a dataset that contains the voting behavior of all Deputies and Senators in all 427 votes of confidence occurred during legislatures XIV-XVIII, from 2001 to 2022. Stenographic record data also provide information on the voting intention of each party, which can be used to construct an indicator variable for whether the MP was elected in a party that supports the government (majority party) or is against the government (opposition party). To determine the party in which the MP was elected, I use the first parliamentary group in the legislature to which they belong. I exclude Senators with a life tenure from the analysis as they continue to receive a parliamentary wage until the end of their life and they never receive a parliamentary pension.[14]

Finally, I have collected data on the annual before-tax income of all Deputies and Senators from 1981 to 2022. Since 1982, Italian elected officials are required to publicly disclose their annual tax returns [Merlo et al., 2010]. As tax returns refer to the previous fiscal year, I have information on each Deputy's and Senator's income in the year prior to entering Parliament. I consider this variable as the data analogue of the private wage w in the model discussed in Section 3. Data from 1981 to 2005 and in 2013-2014 was kindly provided by the Fondazione Rodolfo Debenedetti [2009] and by the Fondazione Openpolis [2017], respectively. For the remaining years, I have digitized the copies of the tax returns of each Deputy and Senator contained in the archive at the 'Servizio Prerogative e Immunità' of the Chamber and the Senate. Only 17 out of 3,032 MPs (0.56 percent) in the analyzed five legislatures have a missing tax return in their first year of parliament, mostly because they started their parliamentary career before 1982 or because their parliamentary career lasted only few months.

[14] A parliamentary group has to have at least 10 members in the Senate or 20 members in the Chamber of Deputies [Camera dei Deputati, 1997; Senato della Repubblica, 2017]. MPs belonging to very small parties are categorized as belonging to the same parliamentary group 'Mixed'. The main results of the analysis are robust to the exclusion of these MPs.

Table 2: Summary statistics

	Pre-reform		Post-reform	
	Tenure below 4.5 years	Tenure above 4.5 years	Tenure below 4.5 years	Tenure above 4.5 years
	A. MP characteristics			
Age	49.18	48.458	46.896	47.107
Female	.109	.124	.325	.293
Number of terms	2.307	2.617	1.409	1.833
Tenure	4.01	4.867	4.003	4.797
High school	.255	.235	.265	.292
Degree	.714	.735	.716	.689
Born in southern Italy	.532	.515	.49	.481
Born in northern Italy	.452	.472	.489	.496
Born outside Italy	.015	.014	.022	.022
Pre-parliament income (€)	114865.5	113827.1	91951.78	99064.87
Private sector	.48	.483	.558	.548
Public sector	.399	.383	.309	.328
Self employed	.093	.106	.113	.105
Out of labor force	.028	.028	.021	.018
Observations	3115	4362	25159	8765
	B. Confidence votes			
Confidence	.643	.646	.749	.632
Abstains/no vote	.126	.211	.208	.209
Vote against party directives	.01	.006	.049	.068
Observations	3115	4362	25159	8765
	C. Legislative bills per day			
Proposed	.00333	.00651	.00277	.0044
Approved	.00039	.00076	.00018	.00019
Ordinary proposed	.00324	.00633	.00262	.00426
Ordinary approved	.00039	.00072	.00015	.00018
Constitutional proposed	.0001	.00042	.00016	.00022
Constitutional approved	0	.00004	.00003	.00001
Observations	167493	161852	627933	333367
	D. Party switches per day			
Any	.0001	.00014	.00043	.00022
To majority	.00001	.00008	.0002	.00005
To opposition	.00001	.00004	.00007	.0001
Observations	167493	161852	627594	333265

Notes: This table shows the sample averages of MP characteristics and outcome variables pre-reform (legislatures XIV-XV) and post-reform (legislatures XVI-XVIII) within one year before and after the 4.5-year tenure threshold. The samples used for *MP characteristics* and *Confidence votes* contain all confidence votes during the analyzed legislatures. The samples used for *Legislative bills* and *Party switches per day* contain all days during the analyzed legislatures. *Legislative bills per day* correspond to the average number of bills proposed or approved in a day by an MP. *Party switches per day* correspond to the probability an MP switches party in a day. Age indicates the age when entering parliament (in years). Tenure indicates the number of years spent in Parliament. Private sector, Public sector, Self employed, and Out of the labor force refers to the MP occupation prior to entering Parliament.

Summary Statistics. The composition of the Italian parliament is similar to parliaments of other European countries with respect to age, gender and education (see Merlo et al. [2010] for a comparison with the US congress). Table 2 contains summary statistics for predetermined characteristics and outcome variables for MPs with parliamentary tenure in a window of one year below and above 4.5 years before and after the reform of the parliamentary pension. Panel A and B show that there were 41,189 confidence votes at the MP level in the 2001-2022 period from legislature XIV to legislature XVIII. Around 70% of these were votes of confidence in favor or against the government (the rest being abstentions or MPs not voting). Mechanically, the average number of parliamentary terms below 4.5 years of tenure (1.51) is lower than the number of terms above (2.09). Likewise, age mechanically differs by approximately 1.5 years. The education level and geographic origin of the MPs do not vary substantially around the threshold, whereas the number of female MPs decreases by 6 percentage points. This is because newly-elected MPs are more likely to be female than reelected MPs. The annual before-tax income in the year prior to entering parliament is almost €$10,000$ lower below 4.5 years of tenure. This is because in the last legislatures there was a large influx of newly elected MPs with very low income (see Figure 3a in Section 6.2). Interestingly, in the year before reaching the tenure threshold the percentage of MPs voting confidence in the government is substantially higher than in the year after (74 vs 64 percent), whereas the probability of voting against party directives is lower (5 vs 6%).

Panel D of Table 2 shows the probability of switching parties by an MP in a day during a window of one year below and above 4.5 years of tenure. We see that the probability of switching to another party is almost twice as high before reaching the tenure requirement than after and this difference is driven uniquely by switches to majority parties.

6 Empirical results

In the main regressions the dependent variable is defined as:

$$Confidence_{ipgt} = \begin{cases} 1 & \text{if MP i elected in party p votes confidence in government g at time t,} \\ 0 & \text{if MP i elected in party p votes no confidence in government g at time t.} \end{cases}$$

Up to the end of legislature XVII, an abstention vote in the Senate counted as a vote against the proposed motion, so I set $Y_{ipgt} = 0$ for these particular cases. In all other cases (abstentions in the Chamber, abstentions in the Senate in legislature XVIII and no vote), I set Y_{ipgt} to missing. Table C8 shows that the decision to abstain or to not vote appears to be negatively affected by the minimum tenure requirement, but the decrease is only marginally significant in one specification.

6.1 Government stability

Table 3 shows the diff-in-disc estimates of a minimum tenure requirement for a parliamentary pension on the probability of voting confidence in the government. In Columns (1)-(3), we see that the effect is positive and significant in all specifications for the entire sample of Deputies and Senators. As expected, the estimates become more precise when we add party-by-government fixed effects and individual MP fixed effects. Exploiting the variation in votes for the same government by the same MP, we can see that the tenure requirement significantly increases the probability of voting confidence in the government by 3 percentage points from an average of 70% MPs voting confidence. This corresponds to 28 additional MPs voting confidence in the government thanks to the parliamentary pension. The estimates with all fixed effects are significantly positive and very similar for the Chamber of Deputies and for the Senate (Table C6).

Table 3: Diff-in-disc estimates of minimum tenure requirement on confidence, all sample.

	(1)	(2)	(3)
Post*Tenure under 4.5 years	0.055***	0.056***	0.032***
	(0.020)	(0.014)	(0.007)
N	33,038	33,038	33,038
R-squared	0.02	0.83	0.95
Average outcome	0.70	0.70	0.70
MP FE	NO	YES	YES
Party-by-gov. FE	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

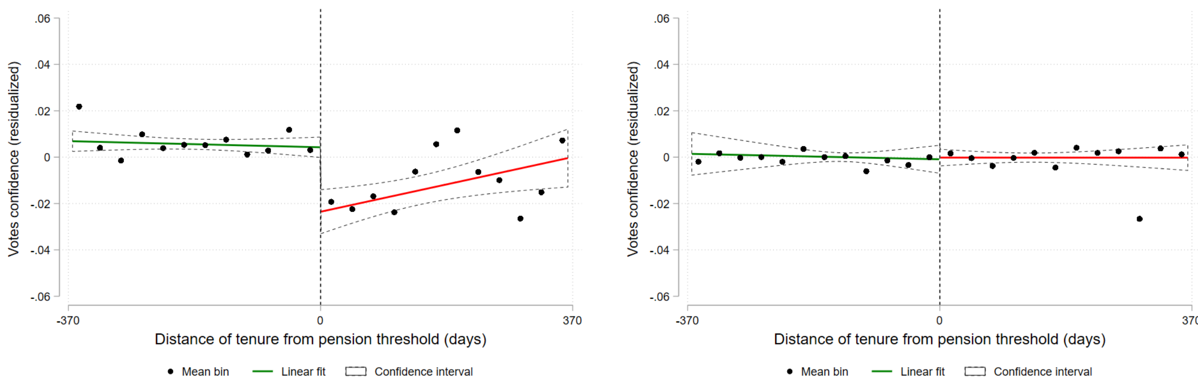
Given that the regressions include MPs' and party-by-government fixed effects, Figure 2 plots the residuals from a regression of the outcome of interest (vote of confidence) on party-by-government fixed effects and individual MP fixed effects in order to net out these fixed effects, as suggested by Lee and Lemieux [2010].[15]

The residualized outcomes are averaged in monthly bins and plotted on the distance (in days) to the 4.5 year-tenure threshold. The figures include the predicted values of a local linear regression of the residualized outcomes on (normalized) tenure, separately for each side of the cutoff. Confidence intervals are constructed on the linear fit, with errors clustered at the MP level. The fitted lines best illustrate the trends in the data and the size of the jump, whereas the binned averages provide a sense of the underlying variability in the data. This exercise is performed for the entire sample (both Houses) and separately for each House, in the period before and after the introduction of the minimum tenure requirement for a parliamentary pension. Figure 2 confirms the qualitative

[15]Controlling for covariates or residualizing the outcome yields the same consistent estimate of the RD parameter of interest as long as the order of the polynomial of the running variable is correctly specified and the covariates are not discontinuous at the threshold Lee and Lemieux [2010]. In this case, the estimates are robust to linear and quadratic specifications and all pre-determined observables are balanced. Reassuringly, Table 3 and Figure 2 produce the same qualitative results.

results of columns (3), (6) and (9) of Table 3. Reassuringly, the positive effect on confidence is large and significant only after the introduction of the minimum tenure requirement whereas it is very close to zero and insignificant in the pre-treatment period.

Figure 2: Regression discontinuities, post-treatment and pre-treatment



(a) Both Houses, post-treatment

(b) Both Houses, pre-treatment

Notes: These figures show the effect of the parliamentary tenure distance from the 4.5 year-cutoff on the MP's probability of voting confidence in the government. The circles are averages across monthly bins on either side of the threshold, while the solid and dashed lines represent the predicted values and confidence intervals of a local linear regression of the outcome on (days of tenure, normalized) and the fixed effects, separately for each side of the cutoff. The bandwidth includes observations within one year from the 4.5 year-cutoff.

Validity tests. Appendix Section E provides a comprehensive description of all validity tests and robustness checks conducted on these results. Table C7 shows that the effects on confidence are very similar when using local quadratic regressions instead of local linear regressions.[16] Figure D1 shows that the magnitude of the effect is significantly positive and remarkably stable when using different bandwidths around the tenure threshold, from two months up to three years on each side.

I test for the presence of manipulation around the 4.5 years cut-off using the test proposed by Frandsen [2017] in the context of regression discontinuity designs with a discrete running variable, as confidence votes are lumpy and scattered over time. The test cannot reject the null hypothesis (p-value$= 0.342$) of absence of manipulation in the distribution of the tenure of MPs in confidence votes. In Table C3, we see that age is the only pre-determined characteristics of the MPs with a statistically significant jump at the threshold. This rules out possible selection effects in a close window around the threshold that could impair the identification strategy.

To assess the possibility that the main result arises from random chance rather than from a causal relationship, I perform a set of diff-in-disc estimations at placebo thresholds below and

[16] The effect is insignificant in the local quadratic regression without any fixed effect, but it becomes significantly positive when adding the fixed effects.

above 4 years and 6 months. Figure D2 shows that all the placebo estimates are lower than the true-threshold coefficient for the confidence vote and the cumulative distribution function is much steeper around 0.

6.2 Party discipline

An analysis of the heterogeneous effects of the policy provides further tests on the predictions of the political-agency model presented in Section 3. In particular, according to Prediction 1, we expect the minimum tenure requirement to have an unambiguously positive impact on the probability to vote confidence for majority MPs, whereas, according to Prediction 2, the effect on opposition MPs is ambiguous, depending on which (if any) of the pivotal and party-discipline effect dominates.

Table 4: Diff-in-disc estimates of minimum tenure requirement on confidence, by type of party.

	Majority party			Opposition party		
	(1)	(2)	(3)	(4)	(5)	(6)
Post*Tenure under 4.5 years	0.048*** (0.011)	0.048*** (0.009)	0.041*** (0.009)	-0.087*** (0.018)	0.000 (0.012)	0.004 (0.011)
N	22,785	22,785	22,785	9,221	9,221	9,221
R-squared	0.01	0.78	0.78	0.04	0.89	0.90
Average outcome	0.96	0.96	0.96	0.10	0.10	0.10
MP FE	NO	YES	YES	NO	YES	YES
Party-by-gov. FE	NO	NO	YES	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

As we can see in Table 4, the model predictions hold. The effect of the tenure requirement is positive and highly significant for majority MPs, whereas it is negative for opposition MPs. The party-discipline effect appears to dominate the pivotal effect, even though the significance of the negative effect disappears when we control for party-by-government and MPs' fixed effects (column 6), as confirmed by Figure D12.[17] Note that according to the model presented in Section 3, these empirical results imply that the minimum tenure requirement is unambiguously distortionary. The minimum tenure requirement unambiguously decreases the expected social surplus when the party-discipline effect dominates.

Table 5 confirms that the minimum tenure requirement increased party discipline. This table reports the estimates of the diff-in-disc coefficient β_0 in (6) when I use as dependent variable an indicator variable equal to 1 if the MP voted against party directives at time t and 0 otherwise.

[17] As we can see in Figure D8, the significantly positive treatment effect on majority members is robust to the choice of different bandwidths, whereas the effect on opposition parties appears to be sensitive to the choice of the bandwidth.

The pension incentive reduced the probability of voting against party directives by 3.4 percentage points, from an average of 5 percentage points.

Table 5: Diff-in-disc estimates of minimum tenure requirement on vote against party directives, all sample.

	(1)	(2)	(3)
Post*Tenure under 4.5 years	-0.042***	-0.035***	-0.027***
	(0.008)	(0.007)	(0.007)
N	39,888	39,888	39,888
R-squared	0.01	0.64	0.67
Average outcome	0.05	0.05	0.05
MP FE	NO	YES	YES
Party-by-gov. FE	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

The effects weaken with majority margins, distance from retirement age, and income According to Prediction 1 and 2, if the party-discipline effect dominates, a larger government majority should weaken the pension incentive. This is because the probability to secure the pension in a first parliamentary term is higher and MPs are less induced to be loyal to their own party, reducing the party-discipline effect. In addition, with larger majorities MPs are less likely to be pivotal voters, reducing the pivotal effect. To test this prediction, I define the government 'majority margin' in each house as the difference between the number of MPs voting confidence in the first vote of confidence (government confirmation) and the minimum number of MPs to obtain a majority (315 in the Chamber and 158 in the Senate). I restrict the sample to the period after the introduction of the minimum tenure requirement and perform the estimation interacting all the regressors in (7) with the majority margin, separately for majority and opposition MPs.[18] These estimations rely on the conditional independence assumption that the interaction is not capturing the effects of correlated unobservables.

Table 6 shows the estimates of the relevant interaction: $D_i \cdot Margin_{it}$. According to the model, a larger majority margin should dampen the effects for both majority MPs and opposition MPs. The empirical estimates in Table 6 validate the model predictions: the estimates of the interaction coefficient are significantly negative for majority MPs and positive for opposition MPs,

[18]Let $Margin_{ig}$ be the majority margin of government g in the first confidence vote in the parliamentary house of MP i. I estimate

$$Y_{ipgt} = \psi_0 + \psi_1 \tilde{Z}_{it} + D_{it}(\omega_0 + \omega_1 \tilde{Z}_{it}) + Margin_{ig}[\psi_2 + \psi_3 \tilde{Z}_{it} + D_{it}(\omega_2 + \omega_3 \tilde{Z}_{it})] + \nu_{ipgt} \quad (7)$$

Table 6 reports the coefficient of interest ω_2, as well as the baseline coefficient ω_0.

except in the last specification in which effects were null in the first place. Interestingly, in the first two specifications for opposition MPs we see that the party discipline effect dominates the pivotal effect even at a zero majority margin as the non-interacted coefficient is significantly negative.

Controlling for party fixed effects, I find that one hundred additional MPs sustaining the government dampens the effect of the pension by 2.8 percentage points on majority MPs and by 4.2 percentage points on opposition MPs. Similarly, Table C15 shows that larger majority margins reduce the tenure-requirement effects on party discipline.

Table 6: Regression discontinuity estimates on confidence, heterogeneity by majority margin.

	Majority party			Opposition party		
	(1)	(2)	(3)	(4)	(5)	(6)
Tenure under 4.5	0.079***	0.074***	0.073***	-0.143***	-0.000	0.012
years	(0.020)	(0.016)	(0.016)	(0.030)	(0.019)	(0.019)
Tenure under 4.5	-0.027***	-0.026***	-0.027***	0.064***	-0.020*	-0.010
years*Majority margin ('00)	(0.010)	(0.008)	(0.008)	(0.016)	(0.011)	(0.009)
N	18,939	18,939	18,939	7,245	7,245	7,245
R-squared	0.01	0.78	0.79	0.04	0.89	0.90
MP FE	NO	YES	YES	NO	YES	YES
Party FE	NO	NO	YES	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. Majority margin indicates the majority margin in the house in which the confidence vote took place measured in hundred MPs. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

If MPs discount the future, we also expect that closeness to retirement age amplifies the effects of the pension incentive.[19] In Tables C11 and C14, I perform the same exercise as above using age when entering parliament as the interaction variable. In line with the model predictions, age of entry amplifies the positive effect on confidence for majority MPs, the negative effect on confidence for opposition MPs, and the negative effects on voting against party directives. Table shows that income tend to decrease the magnitude of the effects, even though the interaction coefficient is insignificant.

Finally, I perform the same heterogeneity analysis over real gross earnings (in 2005 Euros) in the year prior to entering parliament which is reported in the tax returns. Given that income increases with age, I residualize earnings by a quadratic polynomial in age to have a measure of the outside option that is comparable across MPs. In line with model predictions, Tables C12 and C14 show that higher earnings reduce the effects on both confidence and party-discipline, even though the interaction coefficients are insignificant.

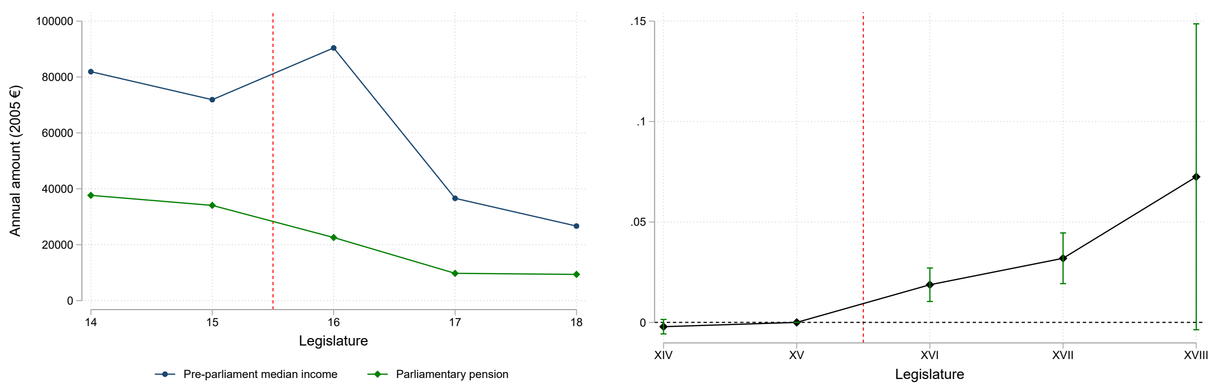
Richer, younger MPs under stable governments tend to be less affected by tenure-dependent

[19] If we extend the retirement age to a general time T, the model predicts that if $\beta < 1$ all the effects are amplified when T is lower (i.e. when the MP is closer to the retirement age).

pensions and vote more independently than poorer, older MPs under unstable governments.[20]

Effects increase over time as private earnings and reelection probabilities decrease. Figure 3b shows the timing of the effect performing the RD regression in Equation (5), separately for each of the five legislatures in the analyzed period. Reassuringly, the effect is not significantly different from zero before legislature XVI and it becomes significantly positive when the minimum tenure requirement is introduced.[21]

Figure 3: Pre-parliament earnings, parliamentary pension and confidence effects by legislature



(a) Median earnings and pension, by legislature

(b) RD coefficients on confidence, by legislature

Notes: Panel (a) shows the median earnings in the year prior to entering parliament and the minimum parliamentary pension they would obtain after 4.5 years of tenure, separately for each legislature. Panel (b) shows the RD coefficient and its 95% confidence interval estimating regression Equation (5), separately for each legislature. The red dashed line indicates the introduction of the minimum pension requirement at 4.5 years.

The effect appears to increase over time, despite the pension amount decreases in the last two legislatures (Figure 3a). The confidence interval is larger in the last legislature because it ended prematurely and there are less observations in the window around the 4.5 year threshold. Two reasons can explain the increase in the point-estimates over the legislatures: a negative shift in the distribution of newly-elected MPs' private income and a decrease in the probability of being elected.

The Italian parliament experienced a drastic turnover between legislature XVI and legislatures XVII-XVIII. The anti-establishment '5-star Movement' elected zero MPs in legislature XVI, but

[20]Another interesting heterogeneity analysis is the effect by gender. Table C10 shows that the effects on voting against party directives are homogeneous across gender. The effect on confidence appears to be weaker for women than for men, even if the difference is not significant in all specifications (Table C9). Estimates for women are more noisy, as the share of female MPs was very low in legislature XIV (11 percent) and only gradually increased up to 36 percent in legislature XVIII.

[21]Legislature XV lasted less than two years and does not have a sufficient number of observations around the tenure threshold of 4.5 years to perform the RD estimation.

became the party with the largest number of MPs in the following two legislatures. Using newly-collected data on MPs' tax returns for the year prior to entering parliament, we can see that the distribution of pre-parliament income changed substantially between legislature XVI and XVII-XVIII. Figure 3a shows that the median pre-parliament annual income (in 2005 €) dropped from 90424€ in legislature XVI to 36613€ in legislature XVII and 26690€ in legislature XVIII. The median age of newly-elected MPs also decreased from 50 years in legislature XVI to 45-46 years in legislatures XVII-XVIII (Figure D14). Controlling for a quadratic polynomial of age when entering parliament, the decrease in average income from legislature XVI to legislatures XVII-XVIII is significant and amounts to about 50,000 euros (Figure D15).

Figure D16 shows that the pre-parliament annual income distribution in legislature XVI first-order stochastically dominates the pre-parliament annual income distributions in legislature XVII and legislature XVIII. The share of MPs earning zero income increased from 0.54 percent in legislature XVI to 8.48 and 6.62 percent in legislature XVII-XVIII, respectively. The share of MPs earning income below the poverty line ($11,239$€ in 2005) increased from 3.78 percent in legislature XVI to 17.78 and 22.68 percent in legislature XVII-XVIII, respectively. The parliamentary pension became a larger share of the pre-parliament median income in legislature XVII-XVIII (27-35 percent) relative to legislature XVI (25 percent). Assuming diminishing marginal utility, the parliamentary pension represented a stronger incentive for MPs with lower private income and lower expected private pension.

An additional reason for the larger increase in legislature XVIII is that the number of MPs in each House was cut by 36.5 percent starting from legislature XIX, thus reducing the probability that MPs would be reelected and would obtain a parliamentary pension in the following legislature. According to the model, a fall in the reelection probability should reduce loyalty to the party by both majority and opposition MPs, resulting in a less positive party-discipline effect for majority MPs and in a less negative party-discipline effect for minority MPs. Figure D6 is in line with the model predictions, with a drop for the effect on majority MPs and an increase for the effect on minority MPs in legislature XVIII. Overall, the pivotal effect which is unambiguously positive for both majority and opposition MPs becomes predominant with respect to the party-discipline effect and the point-estimate increases.

6.3 Party switches

The model presented in Section 3 abstracts away from the possibility that MPs change party during the legislature. But Italian MPs that 'cross the aisle' can continue to hold their seat without having to resign. Switching from a parliamentary group to another is a formal act that is always recorded in the data.

To test whether the parliamentary pension incentive induced MPs to switch party during the legislature, I created a panel dataset at the daily level for each MP who hold a parliamentary seat in legislatures XIV-XVIII. The main outcome variable $switch_{it}$ is a binary variable equal to 1 if MP i switched parliamentary group in day t and 0 if MP i had a seat in Parliament in day t but did not switch parliamentary group.

There are 3,033 MPs that hold a seat for at least one day in legislatures XIV-XVIII for a total of 7,348,110 days spent in Parliament. Less than a third (994) changed parliamentary group at least once during a legislature for a total of 1,543 switches in the analyzed period. This corresponds to a 0.02% probability of an MP switching party in a single day of their parliamentary tenure.

In Figure D17, we see that in legislature XVII and XVIII switches (in particular to majority parties) appear to be more frequent just before 4.5 years since the start of the legislature than after. We do not see a clear pattern in this sense before the introduction of the minimum tenure requirement.

Table 7 and Figure 4 show that the reform significantly increased the probability of switching to another parliamentary group in the year before reaching the tenure requirement than in the year after. The probability to switch a party in a certain day more than doubles from 0.03% to 0.08%. This effect appears to be entirely driven by switches towards majority parties, with a null effect on switches towards opposition parties. There is a significant increase in party switches from opposition to majority parties (Table C16), whereas the number of party switches from majority to opposition is insignificant or marginally significant according to the specification (Table C17).

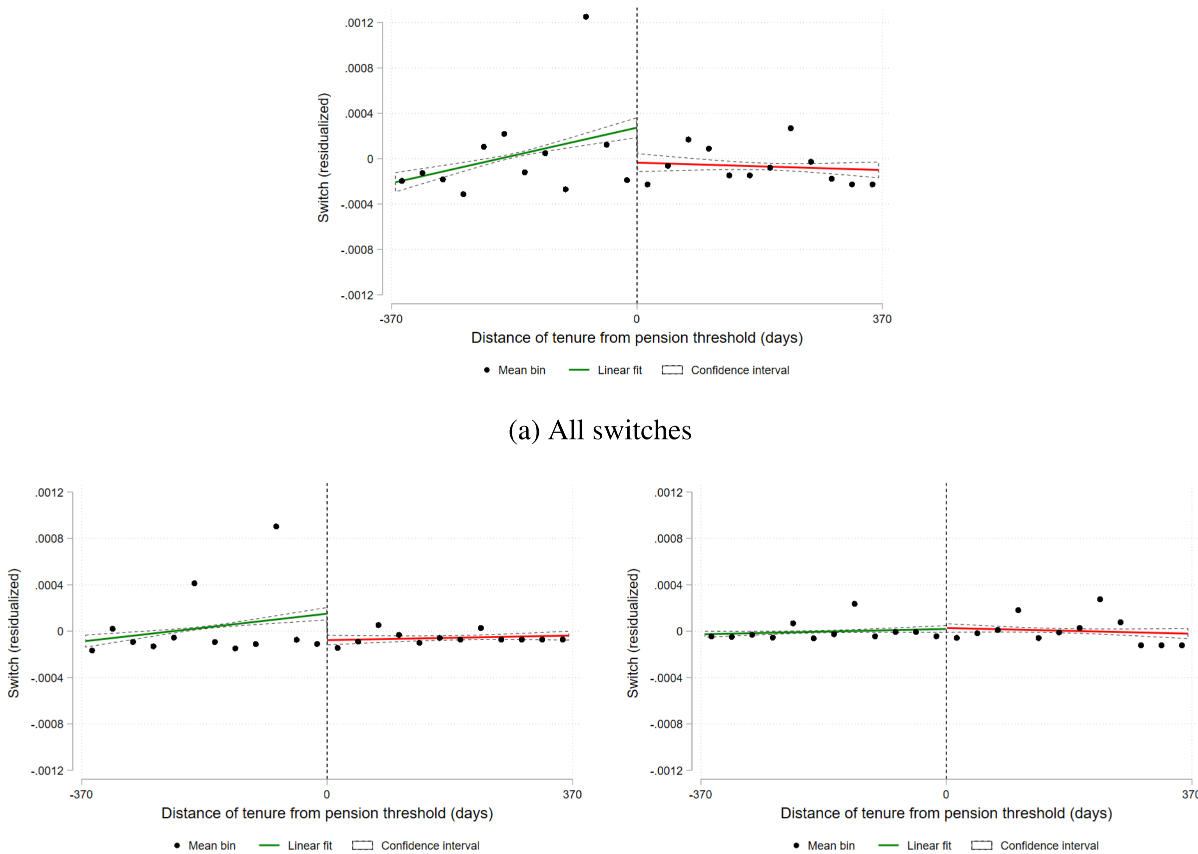
This is consistent with the idea that the tenure requirement incentivized MPs to avoid early elections and obtain a parliamentary pension in the first term, by switching party affiliation. The reelection incentive makes more convenient to switch party altogether rather than defying party directives without changing affiliation.

Table 7: Diff-in-disc estimates of minimum tenure requirement on party-switches, all sample.

	Switch to any party			Switch to majority party			Switch to opposition party		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Post*Tenure under 4.5 years	0.0005***	0.0004***	0.0005***	0.0004***	0.0004***	0.0004***	0.0000	0.0000	0.0000
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0000)	(0.0000)	(0.0000)
N	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204
R-squared	0.0001	0.0030	0.0033	0.0001	0.0021	0.0023	0.0000	0.0027	0.0031
Average outcome	0.0003	0.0003	0.0003	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
MP FE	NO	YES	YES	NO	YES	YES	NO	YES	YES
Party-by-gov. FE	NO	NO	YES	NO	NO	YES	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Figure 4: RD on party switches after the introduction of the minimum tenure requirement



(a) All switches

(b) Switches to majority parties

(c) Switches to opposition parties

Notes: These figures show the effect of the parliamentary tenure distance from the 4.5 year-cutoff on the MP's probability of voting confidence in the government. The circles are averages across monthly bins on either side of the threshold, while the solid and dashed lines represent the predicted values and confidence intervals of a local linear regression of the outcome on (days of tenure, normalized) and the fixed effects, separately for each side of the cutoff. The bandwidth includes observations within one year from the 4.5 year-cutoff.

6.4 Effort

Given that reelections increase the probability of obtaining a parliamentary pension by prolonging time spent in Parliament, tenure-dependent benefits can also be regarded as a reelection incentive. An extension of the political agency model that includes effort predicts that tenure-dependent benefits induce MPs to put more effort into signaling their legislative productivity and boost their chances of party nomination and reelection (Appendix Section B). Empirically, we can measure effort by the probability of submitting a legislative bill as first sponsor in each day during a parliamentary term and the probability that the submitted bill is approved by the Parliament [Ferraz and Finan, 2009]. During the five legislatures analyzed, only 2,343 of the 27,726 legislative proposals

(8.5%) were approved.

Columns (1)-(3) of Table 8 show that there is no significant effect in the probability of submitting a bill. Although drafting a law is a good proxy for MPs' effort, a more impactful measure in terms of voters' welfare is whether the MP submits bills that are eventually approved. Controlling for MPs fixed effects as in Columns (5)-(6) of Table 8, we find that the tenure requirement increases the daily probability that an MP submits a bill that is eventually approved by 0.04-0.05 percentage points from an average of 0.03 percent. In Tables C18 and C19 we see that the probability to submit a law that will be approved increased both for ordinary and constitutional laws, and there is a significant increase in constitutional laws proposals even if this constitutes a small fraction of total bills (6.8%).

If voters value legislative effort, the increase in legislative output represents another way through which tenure-dependent benefits can influence welfare, adding complexity to their overall impact. While these benefits tighten party control, reducing voters' welfare, they also enhance government stability and legislative effort, leading to ambiguous normative implications.

Table 8: Diff-in-disc estimates of minimum tenure requirement on law proposed and approved.

	Law proposed			Law proposed and approved		
	(1)	(2)	(3)	(4)	(5)	(6)
Post*Tenure under 4.5 years	-0.00036	0.00043	0.00042	0.00016	0.00041***	0.00054***
	(0.00047)	(0.00047)	(0.00054)	(0.00013)	(0.00013)	(0.00014)
N	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204
R-squared	0.001	0.009	0.010	0.000	0.005	0.006
Average outcome	0.0037	0.0037	0.0037	0.0003	0.0003	0.0003
MP FE	NO	YES	YES	NO	YES	YES
Party-by-gov. FE	NO	NO	YES	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

6.5 Back-of-the-envelope calculations on government stability

To have a sense of the magnitude of these results, we can perform a back-of-the-envelope calculation on the percentage of confidence votes Italian governments would have lost in the absence of the minimum tenure requirement. Based on column (3) in Table 4, I assume that the effect of the policy was a homogeneous increase of 4.1 percentage points in the confidence votes by majority MPs with a tenure below 4.5 years over the entire post-treatment period. Based on column (6) in Table 4, I assume that the policy did not have any effect on opposition MPs. This exercise hinges on the strong assumptions that the effect is homogeneous across majority MPs (under 4.5 years of tenure) and across opposition MPs and that we can extrapolate the treatment effect far from the tenure threshold, while in fact it is identified only locally.

Since the introduction of the minimum tenure requirement (legislatures XVI-XVIII), there have been 344 confidence votes. According to these calculations, in eight of them (2.3 percent) the government would have lost a vote of confidence had the tenure requirement for a parliamentary pension been absent. Seven of them occurred during the government 'Berlusconi IV' in legislature XVI and one of them during the government 'Conte II' in legislature XVI. Taking these calculations at face value, the government 'Berlusconi IV' would have fallen on December 10, 2010 instead of resigning on November 17, 2011.[22]

It is interesting to note that the spread between the ten-year Italian Treasury Bonds and the German Bund rose sharply from 1.6 percent in June 2011 to 5.5 percent in November 2011, starting to decline only when Berlusconi resigned and was replaced by Mario Monti [Manasse et al., 2013]. Italy experienced a severe sovereign debt crisis in the final months of Berlusconi's government, which an earlier defeat in a confidence vote could have potentially mitigated.

7 Conclusions

This paper employed a difference-in-discontinuities design to test a political agency model of MPs' opportunistic behavior in predicting the impact of the introduction of a minimum tenure requirement to obtain a parliamentary pension in the Italian Parliament. The change in the parliamentary perquisite increases the probability of voting confidence in the government by 3 percentage points from an initial average of 70 percent. The tenure requirement increased confidence votes by MPs elected in parties that support the government, whereas it does not affect confidence votes by MPs' elected in opposition parties, despite an increase in switches towards majority parties during the legislature. Additionally, the policy incentivized legislative effort, as evidenced by an increase in approved bill proposals.

The empirical estimates are consistent with the predictions of a political-agency model in which voters have imperfect information about government performance and MPs are opportunistically interested in reaching the tenure requirement to obtain a parliamentary pension. Beyond the direct incentive to keep the current government in power, the policy discourages voting against party directives: it induces opposition (majority) MPs to vote against (for) the government so as to increase the probability of being reelected and reach the tenure requirement in a second term in case the government falls.

A caveat on these estimates is that the internal validity of the empirical strategy comes at the price of lower external validity, as is always the case in local econometric designs based on policy discontinuities. Understanding how tenure requirements for parliamentary perquisites can affect

[22]The government 'Conte II' resigned after that vote of confidence, so the tenure requirement would have been inconsequential in this case.

the quality of the elected policymakers and their decisions to enter and exit the political career is an interesting avenue for further research.

The main policy insight of this paper is that parliamentary benefits can affect policy decisions on the margin and therefore should be designed carefully. Monetary incentives for legislators to remain in power entail a trade-off between political stability and party discipline. They can reduce the policy uncertainty associated with government crises but they can also become a hidden tool for parties to enhance their control over legislators. Tenure requirements for parliamentary benefits can be distortionary as they induce to follow party directives against voters' interest. This normative result calls for the evaluation of alternative pension schemes, such as the abolition of tenure requirements for parliamentary pensions as in the United Kingdom, or the elimination of special pension schemes for MPs as in France. In the end, MPs could simply earn pension rights on the same terms as the rest of citizens they represent.

References

Acconcia, A. and C. Ronza (2022, May). Women's Representation in Politics and Government Stability. CSEF Working Papers 611, Centre for Studies in Economics and Finance (CSEF), University of Naples, Italy. 4

Alesina, A., S. Özler, N. Roubini, and P. Swagel (1996). Political instability and economic growth. *Journal of economic growth (Boston, Mass.) 1*(2), 189–211. 1

Baker, S. R., N. Bloom, and S. J. Davis (2016, 07). Measuring Economic Policy Uncertainty*. *The Quarterly Journal of Economics 131*(4), 1593–1636. 1

Barro, R. J. (1973). The control of politicians: An economic model. *Public choice 14*(1), 19–42. 4

Besley, T. (2004). Paying politicians: Theory and evidence. *Journal of the European Economic Association 2*(2-3), 193–215. 4

Besley, T. (2007). *Principled Agents?: The Political Economy of Good Government*. The Lindahl Lectures. Oxford: Oxford University Press. 4, 10, 14

Besley, T. and A. Case (1995). Does electoral accountability affect economic policy choices? evidence from gubernatorial term limits. *The Quarterly journal of economics 110*(3), 769–798. 4

Besley, T. and S. Coate (1998). Sources of inefficiency in a representative democracy: A dynamic analysis. *The American economic review 88*(1), 139–156. 14

Bloom, N. (2014, May). Fluctuations in uncertainty. *Journal of Economic Perspectives 28*(2), 153–76. 1

Bloom, N., S. Bond, and J. Van Reenen (2007). Uncertainty and Investment Dynamics. *The Review of Economic Studies 74*(2), 391–415. 1

Bordo, M. D., J. V. Duca, and C. Koch (2016). Economic policy uncertainty and the credit channel: Aggregate and bank level u.s. evidence over several decades. *Journal of Financial Stability 26*, 90–106. 1

Camera dei Deputati (1997). Regolamento della Camera. Parte I. Capo III. Dei Gruppi parlamentari. `http://leg14.camera.it/cost_reg_funz/663/912/928/listaArticoli.asp`. [Online; accessed 29-May-2022]. 20

Camera dei Deputati (2001). XIV legislatura. Deputati. Trattamento economico. https://leg14.camera.it/deputatism/4385/documentotesto.asp. [Online; accessed 29-May-2022]. 9

Camera dei Deputati (2007). XV legislatura. Deputati. Trattamento economico. http://leg15.camera.it/deputatism/4385/documentotesto.asp. [Online; accessed 29-May-2022]. 9

Camera dei Deputati (2013). XVI legislatura. Deputati. Trattamento economico. https://leg16.camera.it/383?conoscerelacamera=4. [Online; accessed 29-May-2022]. 9

Camera dei Deputati (2022). Parliamentary confidence in the government. https://en.camera.it/4?scheda_informazioni=27. [Online; accessed 29-May-2022]. 5, 10

Canen, N., C. Kendall, and F. Trebbi (2020a). Unbundling polarization. *Econometrica 88*(3), 1197–1233. 4

Canen, N. J., C. Kendall, and F. Trebbi (2020b, December). Political parties as drivers of u.s. polarization: 1927-2018. Working Paper 28296, National Bureau of Economic Research. 4

Carozzi, F., D. Cipullo, and L. Repetto (2022). Political fragmentation and government stability: Evidence from local governments in spain. *American economic journal. Applied economics 14*(2), 23–. 4

Caselli, F. and M. Morelli (2004). Bad politicians. *Journal of Public Economics 88*, 759–782. 4

Chiaramonte, A. and R. d'Alimonte (2018, May). The new italian electoral system and its effects on strategic coordination and disproportionality. *Italian Political science 3*(1), 8–18. 10, 11

Congressional Research Service (2023). Retirement Benefits for Members of Congress. https://crsreports.congress.gov/product/pdf/RL/RL30631. [Online; accessed 12-October-2023]. 1, 7, 8, 11

Daconto, C. (2022). Pensioni e vitalizi dei parlamentari: come funzionano. Panorama. 9

De Santis, V. (2020). *Indennità e vitalizi: per uno studio dell'art. 69 della Costituzione.* Studi di Diritto Pubblico. Franco Angeli. 8, 9

Diermeier, D., M. Keane, and A. Merlo (2005). A political economy model of congressional careers. *The American economic review 95*(1), 347–373. 4

Diermeier, D. and A. Merlo (2000). Government turnover in parliamentary democracies. *Journal of economic theory 94*(1), 46–79. 3

Dipartimento per le Riforme Istituzionali (2022). La riduzione del numero dei parlamentari. https://www.riformeistituzionali.gov.it/it/la-riduzione-del-numero-dei-parlamentari. [Online; accessed 29-May-2022]. 5

European Parliament (2021). MEPs' Pension Rights before and after the Members' Statute in 2009. https://www.europarl.europa.eu/thinktank/en/document/IPOL_STU(2021)659763. [Online; accessed 12-October-2023]. 7, 8, 11

Ferejohn, J. (1986). Incumbent performance and electoral control. *Public choice 50*(1/3), 5–25. 4

Ferraz, C. and F. Finan (2009). Motivating politicians: The impacts of monetary incentives on quality and performance. Technical report, National Bureau of Economic Research. 4, 31

Fisman, R., N. A. Harmon, E. Kamenica, and I. Munk (2015). Labor supply of politicians. *Journal of the European Economic Association 13*(5), 871–905. 4

Fondazione Openpolis (2017). Patrimoni trasparenti. Open data. http://patrimoni.openpolis.it/. [Online; accessed 29-May-2022]. 20

Fondazione Rodolfo Debenedetti (2009). Membri del Parlamento Italiano. Dataset su Deputati italiani (1948-2008). https://www.frdb.org/dati/membri-del-parlamento-italiano/. [Online; accessed 29-May-2022]. 20

Frandsen, B. R. (2017). Party bias in union representation elections: Testing for manipulation in the regression discontinuity design when the running variable is discrete. In *Advances in econometrics*, Volume 38, pp. 281–315. Emerald Publishing Limited. 2, 24, 9

Fusani, C. (2007). Stretta su vitalizi e viaggi. Via ai tagli di Camera e Senato. La Repubblica. 9

Gagliarducci, S. and T. Nannicini (2013). Do better paid politicians perform better? disentangling incentives from selection. *Journal of the European Economic Association 11*(2), 369–398. 4

Gagliarducci, S. and M. D. Paserman (2012). Gender interactions within hierarchies: Evidence from the political arena. *The Review of economic studies 79*(3), 1021–1052. 4

Gelman, A. and G. Imbens (2019). Why high-order polynomials should not be used in regression discontinuity designs. *Journal of business economic statistics 37*(3), 447–456. 19

Grembi, V., T. Nannicini, and U. Troiano (2016). Do fiscal rules matter? *American economic journal. Applied economics 8*(3), 1–30. 2, 14, 18

Groseclose, T. and K. Krehbiel (1994). Golden parachutes, rubber checks, and strategic retirements from the 102d house. *American journal of political science 38*(1), 75–99. 4

Hall, R. L. and R. P. Van Houweling (1995). Avarice and ambition in congress: Representatives' decisions to run or retire from the u.s. house. *The American political science review 89*(1), 121–136. 4

House of Commons (2019). Parliamentary Contributory Pension Fund (PCPF) MPs' Pension Scheme (Final Salary Section) Guide for Scheme members. https://www.mypcpfpension.co.uk/cdn/guides/HOP_MP_member_guide-FS_v3.pdf. [Online; accessed 12-October-2023]. 7, 8, 11

Iizuka, T., K. Nishiyama, B. Chen, and K. Eggleston (2021). False alarm? estimating the marginal value of health signals. *Journal of public economics 195*, 104368–. 19

ISTAT (2013). Trattamenti pensionistici e beneficiari. https://www.istat.it/it/archivio/87850. [Online; accessed 29-May-2022]. 9

Keane, M. P. and A. Merlo (2010). Money, political ambition, and the career decisions of politicians. *American economic journal. Microeconomics 2*(3), 186–215. 4

Kolesár, M. and C. Rothe (2018). Inference in regression discontinuity designs with a discrete running variable. *The American economic review 108*(8), 2277–2304. 19

Kotakorpi, K. and P. Poutvaara (2011). Pay for politicians and candidate selection: An empirical analysis. *Journal of public economics 95*(7), 877–885. 4

Lee, D. S. and T. Lemieux (2010). Regression discontinuity designs in economics. *Journal of economic literature 48*(2), 281–355. 19, 23

Manasse, P., G. Trigilia, and L. Zavalloni (2013). Professor Monti and the bubble. https://voxeu.org/article/professor-monti-and-bubble. [Online; accessed 29-May-2022]. 33

Mattozzi, A. and A. Merlo (2008). Political careers or career politicians? *Journal of public economics 92*(3), 597–608. 4

McCrary, J. (2008). Manipulation of the running variable in the regression discontinuity design: A density test. *Journal of Econometrics 142*(2), 698–714. 9

Merlo, A. M., V. Galasso, M. Landi, and A. Mattozzi (2010). Part 1. The Labour Market of Italian Politicians. In T. Boeri, A. Merlo, and A. Prat (Eds.), *The Ruling Class: Management and Politics in Modern Italy*. Oxford University Press. 5, 20, 21

Messner, M. and M. K. Polborn (2004). Paying politicians. *Journal of public economics 88*(12), 2423–2445. 4

Office of the Superintendent of Financial Institutions Canada (2019). Actuarial Report on the Pension Plan for the Members of Parliament. https://www.osfi-bsif.gc.ca/Eng/oca-bac/ar-ra/mp-plm/Pages/MP19.aspx. [Online; accessed 12-October-2023]. 7, 8, 11

Osservatorio sui Conti Pubblici Italiani (2021). Il sistema pensionistico dei parlamentari: possibili implicazioni per la durata della legislatura. https://osservatoriocpi.unicatt.it/ocpi-pubblicazioni-il-sistema-pensionistico-dei-parlamentari-possibili-implicazioni-per-la-durata-della. [Online; accessed 29-May-2022]. 9

Persson, T. and G. Tabellini (2000). *Political economics : explaining economic policy*. Zeuthen lecture book series. Cambridge, Mass.: MIT Press. 1

Preece, D. C., D. J. Mullineaux, G. Filbeck, and S. A. Dennis (2004). Agency theory and the house bank affair. *Review of financial economics 13*(3), 259–267. 4

Rizzo, S. (2018). Vitalizi d'oro: il vero privilegio sono le doppie pensioni. La Repubblica. 7

Rubabshi-Shitrit, A. and S. Hasson (2022). The effect of the constructive vote of no-confidence on government termination and government durability. *West European Politics 45*(3), 576–590. 5, 6

Senato della Repubblica (2017). Il Regolamento del Senato.Capo IV. Dei Gruppi parlamentari. https://www.senato.it/istituzione/il-regolamento-del-senato/capo-iv/articolo-14-1. [Online; accessed 29-May-2022]. 20

Senato della Repubblica (2022a). La Costituzione. Articolo 94. https://www.senato.it/istituzione/la-costituzione/parte-ii/titolo-iii/sezione-i/articolo-94. [Online; accessed 29-May-2022]. 5

Senato della Repubblica (2022b). XVIII legislatura. Senatori. Trattamento economico. https://www.senato.it/leg18/1075?voce_sommario=61. [Online; accessed 29-May-2022]. 9

Sesto, M. (2021). Pensioni degli onorevoli, perché il 24 settembre 2022 è il D day della legislatura. Il Sole 24 ore. 9

World Bank (2023). Life expectancy at birth, total (years) - Italy. https://data.worldbank.org/indicator/SP.DYN.LE00.IN?locations=IT. [Online; accessed 13-Sep-2023]. 7

Appendix A Model appendix

This section provides the proofs of Propositions 1 and 2 of Section 3. Suppose there is no minimum tenure requirement and MPs receive a pension in period 3 independently of the time they spend in Parliament. The expected utility for an MP elected in period 1 in a majority party is:

$$\begin{aligned} V_M^{Pre}(c_{M1}) =& u(\mathcal{I} + \alpha[s_1 c_{M1} + (1-s_1)(1-c_{M1})]\Omega) + \\ &+ \beta\{\pi_M(c_{M1})u(\mathcal{I}+\alpha\Omega) + [1-\pi_M(c_{M1})]u(w)\} + \beta^2 u(\rho + \xi w) \end{aligned} \tag{8}$$

If the government is doing well ($s_1 = 1$), the newly-elected majority MP always votes confidence in the government as:

$$u(\mathcal{I}+\alpha\Omega) + \beta\Delta\pi_M[u(\mathcal{I}+\alpha\Omega) - u(w)] \geq u(\mathcal{I}) \tag{9}$$

given that $\Delta\pi_M \geq 0$.

If the government is not doing well ($s_1 = 0$), the newly-elected majority MP would vote confidence and go against voters' interest if

$$u(\mathcal{I}) + \beta\Delta\pi_M[u(\mathcal{I}+\alpha\Omega) - u(w)] \geq u(\mathcal{I}+\alpha\Omega) \tag{10}$$

Intuitively, MPs follow party directives and vote against voters' interest only if the benefit of party loyalty in terms of reelection probabilities exceeds the utility they get from altruistically benefiting voters in the first term.

Suppose now that a minimum tenure requirement is imposed. The utility function for an MP elected in period 1 in a majority party is:

$$\begin{aligned} V_M^{Post}(c_{M1}) =& u(\mathcal{I} + \alpha[s_1 c_{M1} + (1-s_1)(1-c_{M1})]\Omega) + \\ &+ \beta\{\pi_M(c_{M1})u(\mathcal{I}+\alpha\Omega) + [1-\pi_M(c_{M1})]u(w)\} + \beta^2 u(w) \\ &+ \beta^2\{\sigma(c_{M1}) + [1-\sigma(c_{M1})]\pi_M(c_{M1})\}[u(\rho+\xi w) - u(w)] \end{aligned} \tag{11}$$

Again, if the government is doing well ($s_1 = 1$), the newly-elected majority MP always votes confidence in the government as:

$$\begin{aligned} &u(\mathcal{I}+\alpha\Omega) + \beta\Delta\pi_M[u(\mathcal{I}+\alpha\Omega) - u(w)] \\ &+ \beta^2\{\Delta\sigma[1-\pi_j(0)] + \Delta\pi_j[1-\sigma(1)]\}[u(\rho+\xi w) - u(\xi w)] \geq u(\mathcal{I}) \end{aligned} \tag{12}$$

If the government is not doing well ($s_1 = 0$), the newly-elected majority MP would vote confidence and go against the voters' interest if

$$\begin{aligned}&u(\mathcal{I})+\beta\Delta\pi_M[u(\mathcal{I}+\alpha\Omega)-u(w)]\\&+\beta^2\{\Delta\sigma[1-\pi_j(0)]+\Delta\pi_j[1-\sigma(1)]\}[u(\rho+\xi w)-u(\xi w)]\geq u(\mathcal{I}+\alpha\Omega)\end{aligned}\tag{13}$$

Comparing inequalities (10) and (13), the minimum tenure requirement for the parliamentary pension increases the utility of newly-elected majority MPs for voting confidence in the government against voters' interest by $\beta^2\{\Delta\sigma[1-\pi_j(0)]+\Delta\pi_j[1-\sigma(1)]\}[u(\rho+\xi w)-u(\xi w)]>0$.

Note that the minimum tenure requirement is a distortionary incentive for majority MPs because it induces them to vote in favor of the government, when it would be in the voters' interest to vote against.

In absence of a minimum tenure requirement, the expected utility for an MP elected in period 1 in an opposition party is:

$$\begin{aligned}V_m^{Pre}(c_{m1})=&u(\mathcal{I}+\alpha[s_1c_{m1}+(1-s_1)(1-c_{m1})]\Omega)+\\&+\beta\{\pi_m(c_{m1})u(\mathcal{I}+\alpha\Omega)+[1-\pi_m(c_{m1})]u(w)\}+\beta^2u(\rho+\xi w)\end{aligned}\tag{14}$$

If the government is not doing well ($s_1=0$), the newly-elected opposition MP never votes confidence in the government as:

$$u(\mathcal{I})\leq u(\mathcal{I}+\alpha\Omega)-\beta\Delta\pi_m[u(\mathcal{I}+\alpha\Omega)-u(w)]\tag{15}$$

given that $\Delta\pi_m\leq 0$.

If the government is doing well ($s_1=1$), the newly-elected opposition MP would vote confidence in the voters' interest if

$$u(\mathcal{I}+\alpha\Omega)\geq u(\mathcal{I})-\beta\Delta\pi_m[u(\mathcal{I}+\alpha\Omega)-u(w)]\tag{16}$$

Intuitively, opposition MPs vote according to voters' interest if the altruistic utility of benefiting voters in the first term is higher than their personal gain in terms of reelection probabilities when they follow party directives.

Suppose now that a minimum tenure requirement is imposed. For opposition MPs the pivotal and party-discipline effects have opposite signs. On one hand, opposition MPs would like to vote confidence because if the government wins they would obtain the pension. On the other hand, opposition MPs are afraid to vote confidence because if the government loses, they would be less likely to be reelected and to secure the pension in a second term.

The utility function for an MP elected in period 1 in an opposition party in presence of a minimum tenure requirement is

$$V_m^{Post}(c_{m1}) = u(\mathcal{I} + \alpha[s_1 c_{m1} + (1-s_1)(1-c_{m1})]\Omega) + \\ + \beta\{\pi_m(c_{m1})u(\mathcal{I}+\alpha\Omega) + [1-\pi_m(c_{m1})]u(w)\} + \beta^2 u(w) \\ + \beta^2\{\sigma(c_{m1}) + [1-\sigma(c_{m1})]\pi_m(c_{m1})\}[u(\rho+\xi w) - u(w)] \tag{17}$$

With the minimum tenure requirement, the newly-elected opposition MP might vote confidence even if the government is not doing well ($s_1 = 0$). This occurs when

$$u(\mathcal{I}) + \beta\Delta\pi_M[u(\mathcal{I}+\alpha\Omega) - u(w)] \\ + \beta^2\{\Delta\sigma[1-\pi_j(0)] + \Delta\pi_j[1-\sigma(1)]\}[u(\rho+\xi w) - u(\xi w)] \geq u(\mathcal{I}+\alpha\Omega) \tag{18}$$

A necessary condition for this to occur is that the pivotal effect dominates the party-discipline effect: $\Delta\sigma[1-\pi_j(0)] > \Delta\pi_j[1-\sigma(1)]$. If the party-discipline effect dominates, newly-elected opposition MPs never vote confidence in a bad government, with or without a tenure requirement.

If the government is doing well ($s_1 = 1$), the newly-elected opposition MP would vote confidence in the voters' interest if

$$u(\mathcal{I}+\alpha\Omega) + \beta\Delta\pi_M[u(\mathcal{I}+\alpha\Omega) - u(w)] \\ + \beta^2\{\Delta\sigma[1-\pi_j(0)] + \Delta\pi_j[1-\sigma(1)]\}[u(\rho+\xi w) - u(\xi w)] \geq u(\mathcal{I}) \tag{19}$$

Comparing inequalities (16) and (19), the minimum tenure requirement for the parliamentary pension has an ambiguous effect on the behavior of newly-elected opposition MPs. If the incentive to increase the chances of a government victory to immediately obtain the pension right (*pivotal effect*) is lower than the fear of losing the possibility of being reelected and obtain the pension later in case of government defeat (*party-discipline effect*) ($\Delta\sigma[1-\pi_j(0)] \leq \Delta\pi_j[1-\sigma(1)]$), then the minimum tenure requirement incentives newly-elected opposition MPs to vote *against* the confidence.

Note that if the party-discipline effect dominates the pivotal effect (i.e. the tenure requirement reduces the number of opposition MPs voting confidence in the government), the tenure requirement incentive is distortionary. It incentivizes opposition MPs to vote against the government, even though it would be in the voters' interest to vote in favor.

Appendix B Model extension I. When government stability is positively valued

Suppose the model is the same as in Section 3, but now voters attribute a positive value to government stability and receive utility ψ if the MP votes in favor of the government. Assume that the preference for good governance is stronger than the preference for government stability, that is $\Omega \geq \psi$.

MPs elected in a party $j \in \{m, M\}$ in period 1 choose their confidence vote c_{j1} under state of the world s_1 to maximize expected lifetime utility:

$$\begin{aligned}\max_{c_{j1}} V_j(c_{j1}) &= u(\mathcal{I} + \alpha\{[s_1 c_{j1} + (1-s_1)(1-c_{j1})]\Omega + \psi\}) \\ &\quad + \beta\{\pi_j(c_{j1}) Eu(\mathcal{I} + \alpha[\Omega + s_2\psi]) + [1-\pi_j(c_{j1})]u(w)\} \\ &\quad + \beta^2\{\gamma_j(c_{j1})u(\rho + \xi w) + [1-\gamma_j(c_{j1})]u(\xi w)\}\end{aligned} \tag{20}$$

where $Eu(\cdot)$ is expected utility over the state of the world in the second period s_2.

The expected social surplus generated by MP elected in party $j \in \{m, M\}$ is:

$$\begin{aligned}E(W_j) &= Pr(s_1 = 0)\Omega + \beta\pi_j(0)[\Omega + Pr(s_2 = 1)\psi] \\ &\quad + Pr(s_1 = 1)Pr(c_{j1} = 1|s_1 = 1)\{\Omega + \beta\Delta\pi_j[\Omega + Pr(s_2 = 1)\psi]\} \\ &\quad - Pr(s_1 = 0)Pr(c_{j1} = 1|s_1 = 0)\{\Omega - \psi - \beta\Delta\pi_j[\Omega + Pr(s_2 = 1)\psi]\}\end{aligned} \tag{21}$$

The normative implications of Section 3 remain unchanged if the preference for good governance is sufficiently stronger than the preference for government stability. Formally, the additional required assumption is that $\Omega > \frac{\psi[1+\Delta\pi_j Pr(s_2=1)]}{1-\Delta\pi_j}$.

Again, majority MPs always vote confidence when the government performs well, independently of the tenure requirement: $Pr(c_{M1} = 1|s_1 = 1) = 1$. Since the tenure requirement induces majority MPs to vote confidence: it increases $Pr(c_{M1} = 1|s_1 = 0)$. Therefore, if $\Omega > \frac{\psi[1+\Delta\pi_j Pr(s_2=1)]}{1-\Delta\pi_j}$, the tenure requirement lowers the expected social surplus generated by majority MPs.

Without the tenure requirement, opposition MPs always vote no confidence when the government is not doing well: $Pr(c_{m1} = 1|s_1 = 0) = 0$. The tenure requirement induces opposition MPs to vote no confidence when the party-discipline effect dominates: $Pr(c_{m1} = 1|s_1 = 1)$ decreases. Therefore, the tenure requirement also lowers the expected social surplus produced by opposition MPs if the party-discipline effect dominates.

To conclude, when government stability is positively valued by voters, the minimum tenure requirement decreases ex-ante social surplus if voters have a sufficiently strong preference for good governments relative to stable governments.

Appendix C Model extension II. When effort increases the probability of reelection

Assume now that after voting in the confidence vote, MPs exercise legislative effort e_{jt}. The probability of reelection $\pi_j(c_{j1}, e_{j1})$ is an increasing and concave function of effort in the first period e_{j1}, but effort costs $\kappa(e_{j1})$ where $\kappa(\cdot)$ is a convex and increasing function.[23]

Given confidence vote c_{j1}, an MP of party j chooses effort e_{j1} to maximize expected lifetime utility:

$$\begin{aligned}\max_{e_{j1}} V_j(e_{j1}) =& u(\mathcal{I} + \alpha[s_1 c_{j1} + (1-s_1)(1-c_{j1})]\Omega - \kappa(e_{j1}))+ \\ & \beta\{\pi_j(c_{j1}, e_{j1})u(\mathcal{I} + \alpha\Omega) + [1-\pi_j(c_{j1}, e_{j1})]u(w)\}+ \\ & \beta^2\{\gamma_j(c_{j1}, e_{j1})u(\rho + \xi w) + [1-\gamma_j(c_{j1}, e_{j1})]u(\xi w)\}\end{aligned} \tag{22}$$

Before the reform the probability of obtaining the parliamentary pension is $\gamma_j^{Pre}(c_{j1}, e_{j1}) = 1$. After the reform, the probability of obtaining the parliamentary pension becomes $\gamma_j^{Post}(c_{j1}, e_{j1}) = \sigma(c_{j1}) + [1-\sigma(c_{j1})]\pi_j(c_{j1}, e_{j1})$ which is now a function of effort through the reelection incentive.

The first-order condition with respect to effort can be rearranged as:

$$\begin{aligned}&\frac{u'(\mathcal{I} + \alpha[s_1 c_{j1} + (1-s_1)(1-c_{j1})]\Omega - \kappa(e_{j1}))\frac{\partial \kappa(e_{j1})}{\partial e_{j1}}}{\frac{\partial \pi_j(c_{j1}, e_{j1})}{\partial e_{j1}}} = \\ &\beta\{u(\mathcal{I} + \alpha\Omega) - u(w) + \beta[u(\rho + \xi w) - u(\xi w)][1-\sigma(c_{j1})] \cdot \textit{Post}_t\}\end{aligned} \tag{23}$$

where *Post* is an indicator function equal to 1 if after the reform and 0 otherwise. The numerator of the left-hand side is positive and increasing in effort as the marginal utility is positive and decreasing, effort enters negatively the marginal utility, and the cost function is convex. The denominator of the left-hand side is positive and decreasing in effort as the probability of reelection is an increasing, concave function of effort. Accordingly, the left-hand side is an increasing function of effort. Given that $\beta[u(\rho + \xi w) - u(\xi w)][1-\sigma_j(c_{j1})] > 0$, the right-hand side is higher after the reform (when $Post_t = 1$) relative to before. Therefore, *for a given confidence vote choice* c_{j1}, the introduction of the tenure requirement unambiguously increases the MP's effort. This effect

[23] Given that the MP retires in period 3, in period 2 the MP chooses to exercise zero effort.

should be stronger the higher is the parliamentary pension, the poorer the MP, and the lower the probability of government survival in the first term. However, Section 3 shows that the parliamentary reform affects the confidence vote decision c_{j1} as well, which in turn affects the probability of reelection $\pi_j(c_{j1}, e_{j1})$ and the marginal utility in the first period. The interaction between the two effects makes the impact of the reform on effort ambiguous. If the tenure-requirement induces a change in the confidence vote that makes the probability of reelection independent of the level of effort ($\frac{\partial \pi_j(c_{j1}, e_{j1})}{\partial e_{j1}} \to 0$), the MPs will not exercise any legislative effort and the impact on effort can potentially be negative.

Appendix D Voting against party directives

In this Section, I show that voting against party directives (i.e. voting confidence when elected in an opposition party and voting no confidence when elected in a majority party) is negatively correlated with the probability of being reelected. I restrict to the sample of MPs in their first term and regress:

$$reelected_i = \alpha + \boldsymbol{\beta}\mathbf{x}_i + \delta Deviate_{it} + \eta_{pg} + \varepsilon_{ipgt} \tag{24}$$

where $reelected_i$ is an indicator variable equal to 1 if the MP is ever reelected for a second term in Parliament and 0 otherwise, and $Deviate_{it}$ is an indicator variable equal to 1 if the first-term MP voted against party directives at time t and 0 otherwise. η_{pg} are party-by-government fixed effects and $\mathbf{x}_i$ is a vector of individual characteristics: gender, high school diploma, university degree, born in South, born in Center, foreign, pre-parliament income. I repeat this regression with and without controls and fixed effects. As we can see in Table C1, first-term MPs who vote against their party directives are 10-25 percent less likely to be reelected, depending on the specification.[24] This significantly negative correlation corroborates the corresponding assumption of the model presented in Section 3.

[24] Results do not qualitatively change if I use Probit or Logit models instead of a linear probability model

Table C1: Correlation between voting against party directives and being re-elected in parliament.

	(1)	(2)	(3)
Vote against party directives	-0.105*** (0.028)	-0.137*** (0.039)	-0.063* (0.033)
N	192,817	112,408	112,408
R-squared	0.00	0.02	0.32
Average outcome	0.65	0.65	0.65
Party-by-gov. FE	NO	NO	YES

Notes: The regression sample is restricted to confidence votes by MPs in the first term. Controls are: gender, age when entering parliament, high school diploma, university degree, born in South, North, foreign, pre-parliament income, being employed in the private sector, in the public sector, being self-employed, being out of the labor force prior to entering Parliament. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

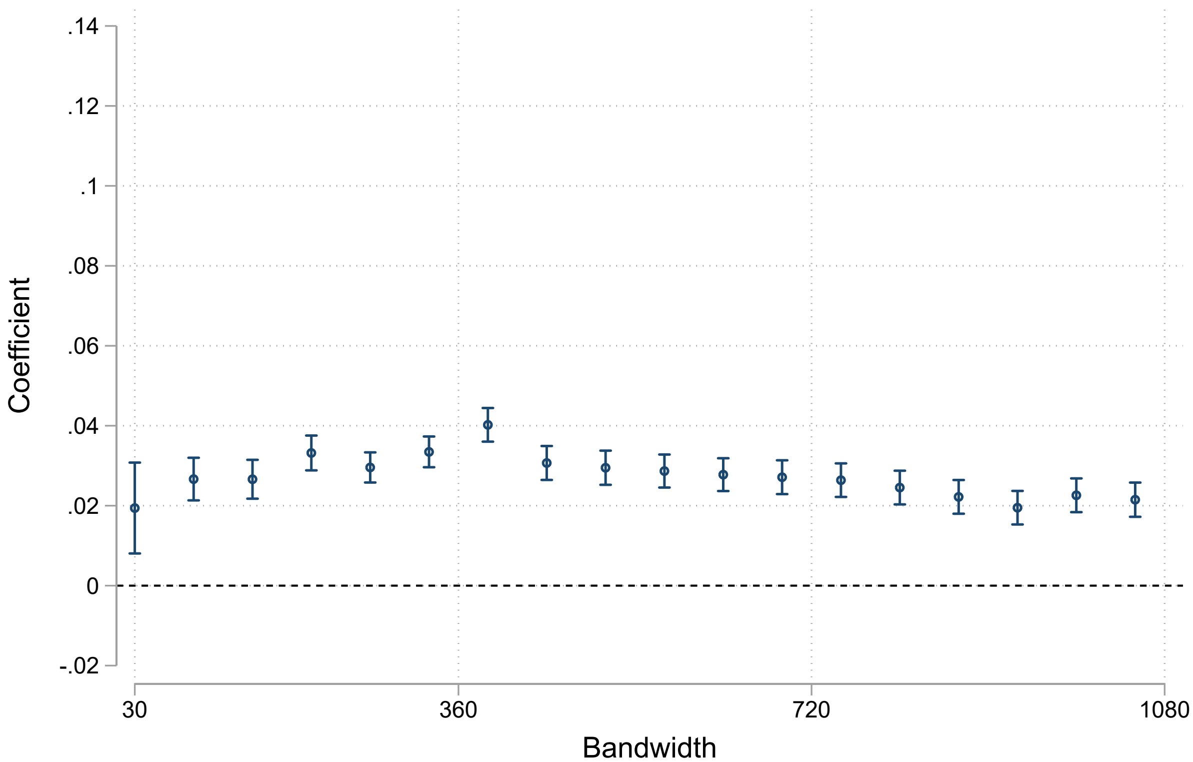
Appendix E Validity tests

To confirm the validity of the empirical strategy, I perform a series of diagnostic tests. Assumption 2 requires that the effect of the severance pay discontinuity at the threshold does not vary with time nor with the pension amount or the electoral law. The severance pay increase does not seem to play a relevant role considering that the effect is not significantly different from zero prior to the introduction of the 4.5-year tenure requirement (Figure 3b). This is understandable considering that the bonus for an extra year of tenure consists only in a moderate increase of the one-off severance payment equal to 80 percent of the last wage. This corresponds to €$8,348$ in legislature XVI which represents only 16.8 percent of the loss in pension contributions (€$49,587$) or four monthly pension payments.[25] Reassuringly, the effect becomes significantly positive in the first legislature in which the minimum tenure requirement was introduced.

Secondly, Figure D1 shows that the magnitude of the effect for the entire Parliament is not sensitive to the bandwidth chosen. The diff-in-disc coefficient is significantly positive and remarkably stable when using different bandwidths around the tenure threshold, from two months up to three years on each side. Figure D7 confirms that the effect for the Senate is robust to the choice of bandwidth. The estimates for the Chamber of Deputies are always positive and significant, except when the bandwidth is very small (two months).

[25] Pension payments were expected to last for sixteen years in 2008, given that life expectancy in Italy was 81 [World Bank, 2023]

Figure D1: Diff-in-disc coefficients: bandwidth sensitivity



Notes: The graph shows the point estimate and the 95% confidence interval of the Diff-in-Disc coefficients on the entire sample (both Houses), estimating the regression specified in Equation (6) using different bandwidths. The horizontal axis shows the number of tenure days from the 4.5-year-cutoff in each side of the bandwidth.

Identification of the treatment effect requires that the MPs or the government do not manipulate the date of the confidence votes to exploit the incentive of the minimum tenure requirement for the parliamentary pension.[26] Manipulation of the timing of confidence votes may occur for two reasons. First, motions of no-confidence may be postponed after 4 years and 6 months of a parliament term in order to let newly-elected MP secure a parliamentary pension and be more willing to vote against the government. This is not the case: there are only two motions of no-confidence in the analyzed period (one in 2010 by the Chamber and one in 2014 by the Senate) and they are both before the 4.5-year threshold. Second, the government may anticipate confidence votes to speed up the legislative procedure before the 4.5-year threshold, trying to exploit the economic incentive for MPs to prolong the legislature. This does not seem to be the case. Figure D4 shows that there is no bunching of confidence votes before the tenure cutoff at 0. There is some bunching around

[26]In a diff-in-disc design, this requirement can be relaxed: it is sufficient that the manipulation of the running variable does not increase over time. However, the absence of any effect in the pre-treatment period allows us to concentrate in the post-period as in a standard RD.

40 days after the cutoff. However, this is only due to the fact that the government called for a vote of confidence on 5 different articles of the same law in the Senate (DDL 2941) in October 25, 2017 and for a vote of confidence on 3 different articles of the same law (Atto Camera 2352) in October 11, 2017. If anything, the government should have anticipated those votes before the threshold to get advantage of the minimum tenure requirement for a parliamentary pension.

When the running variable has a finite number of fixed support points as in this setting (tenure only change in increments of one day and there are several mass points), the standard McCrary [2008] test can falsely reject the null of no manipulation at an excessively high rate or can fail to detect actual anomalies in the running variable's density [Frandsen, 2017]. Therefore, I test for the presence of manipulation around the 4.5 years cut-off using the test proposed by Frandsen [2017] in the context of regression discontinuity designs with a discrete running variable. This test relies only on support points at and immediately adjacent to the RD threshold when the running variable is discrete. The test cannot reject the null hypothesis (p-value= 0.342) of absence of manipulation in the distribution of the tenure of MPs in confidence votes, using a value for the test parameter as low as $k = 10^{-8}$ (p-value=0.342).[27]

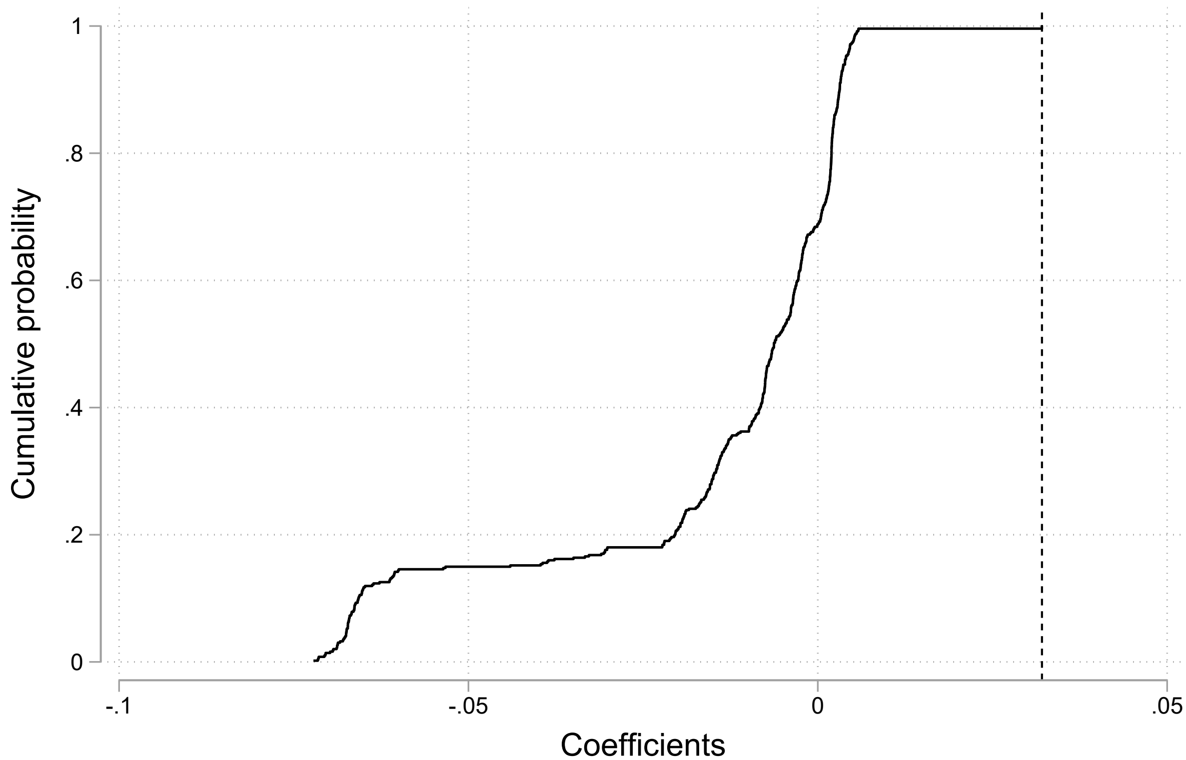
Table C3 further evaluates the absence of manipulation. I implement diff-in-disc estimations with pre-determined variables characteristics (gender, education, birth place, foreign, pre-parliament income) as outcome variables without adding any fixed effects. No pre-determined characteristics show a statistically significant jump at the threshold for the entire sample and for the sample of Deputies. Only one pre-determined characteristic (gender) shows a marginally significant decrease at the threshold for Senators.

To assess the possibility that the main result arises from random chance rather than from a causal relationship, I perform a set of diff-in-disc estimations at placebo thresholds below and above 4 years and 6 months. I place the placebo thresholds at every day from 3 years and 8 months to 4 years and 4 months and at every day from 4 years and 8 months to 5 years and 4 months, in order to stay sufficiently away from the policy thresholds of 3.5, 4.5 and 5.5 years at which the severance payment, the pension age and the pension payment change discontinuously. At these false thresholds, we should not find treatment effects similar to the estimate at the true threshold. Figure D2 shows the cumulative density function of these 488 placebo point estimates obtained using local linear regressions. All these placebo estimates are lower than the true-threshold coefficient for the confidence vote and the cumulative distribution function is much steeper around 0. This placebo test provides evidence that the main result is not driven by mere random noise in the

[27] I implement the test using the Stata command *rddisttestk* [Frandsen, 2017]. The parameter k determines the maximal degree of nonlinearity in the probability mass function that is considered to be compatible with absence of manipulation. A high k allows the mass at the cutoff to deviate substantially from linearity before the test can reject with high probability, whereas a low k means that even with small deviations from linearity the test will reject with high probability. A higher k implies a lower power of the test to detect manipulation.

data.

Figure D2: Placebo estimates



Notes: This graph shows the cumulative distribution of Diff-in-Disc estimates on votes of confidence for both Houses, from placebo local linear regressions in which the cutoff is set in different parts of the tenure distribution. Estimates are computed using the regression in Equation (6) within a 1-year bandwidth. Cut-offs are located at every day from 4 years and 8 months to 5 years and 4 months, in order to stay sufficiently away from the policy thresholds of 3.5, 4.5 and 5.5 years at which the severance payment, the pension age and the pension payment change discontinuously. The vertical dashed line shows the coefficient estimated using the true 4.5-year tenure threshold.

Appendix F Additional Tables

Table C2: Minimum tenure requirements for a parliamentary pension by country in 2009

Country	Pension scheme	Minimum employment in any job	Minimum service as MP or federal employee	Minimum service as MP	Retirement age
United States	Specific		5 years		62
Canada	Specific		6 years		55
EU	Specific			1 year	63
Austria	General	15 years			60- 65
Belgium	Specific			1 month	55
Bulgaria	General	34 years			60- 63
Cyprus	Specific			4 years	60
Czech Republic	General	15 years			56-62
Denmark	Specific			1 year	60
Estonia	General	15 years			60.5- 63
Finland	Specific			1 month	65
France	General	40 years			60
Germany	Specific			1 year	67
Greece	Specific			5 years	65
Hungary	General	15 years			62
Ireland	Specific			3 years	65
Italy	Specific			4.5 years	65
Latvia	General	10 years			62
Lithuania	General	15 years			60- 62.5
Luxembourg	Specific			10 years	65
Malta	Specific			5.5 years	65
Netherlands	Specific		5 years		65
Poland	General	20 years			60- 65
Portugal	General	15 years			65
Romania	General	11 years			58- 63
Slovakia	General	15 years			56-62
Slovenia	General	15 years			58
Spain	General	15 years			65
Sweden	Specific			6 years	65
United Kingdom	Specific			1 month	65

Notes: Pension scheme is 'general' if MPs earn pension rights on the same terms as the rest of the labour market, and 'specific' if MPs earn pension rights in a special scheme for high-level civil servants or MPs only. *Minimum employment in any job* is the minimum number of years that MPs have to be employed across all occupations in their lifetime to obtain a pension in their general pension scheme. *Minimum service as MP or federal employee* refers to the minimum length of actual service as MP or as federal-public employee to obtain a parliamentary pension. *Minimum service as MP* refers to the minimum length of actual service as MP to obtain a parliamentary pension. The sources of this data are the Congressional Research Service [2023], the Office of the Superintendent of Financial Institutions Canada [2019], the European Parliament [2021] and the House of Commons [2019].

Table C3: Diff-in-disc estimates of minimum tenure requirement on pre-determined variables, all MPs.

	(1) Age	(2) Female	(3) High school	(4) Degree	(5) South	(6) North	(7) Foreign	(8) Income (€)	(9) Private sector	(10) Public sector	(11) Self employed	(12) Out of labor force
Post*Tenure under 4.5 years	-0.355 (0.336)	-0.023 (0.014)	-0.002 (0.017)	0.013 (0.018)	0.023 (0.018)	-0.027 (0.018)	0.003 (0.005)	-6913.729 (6160.412)	0.020 (0.022)	-0.035 (0.021)	0.016 (0.012)	-0.001 (0.006)
N	33,038	33,038	21,784	21,784	33,038	33,038	33,038	32,953	20,421	20,421	20,421	20,421
R-squared	0.09	0.03	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00
Average outcome	47.46	0.28	0.27	0.71	0.49	0.49	0.02	91532.43	0.54	0.34	0.11	0.02
MP FE	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
Party-by-gov. FE	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO

Notes: The regression sample is restricted to votes of confidence in a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. Age indicates the age when entering parliament for the first time (in years). No diploma, High school and University degree refer to the highest education level achieved. South, North and Foreign refer to the birthplace. Income refers to the private income in the year prior to entering Parliament and is available for MPs that enter Parliament later than 1981. Public sector, Private sector, Self employed and Out of the labor force refers to the MP occupation prior to entering Parliament. Data on education levels is not available for senators. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Table C4: Diff-in-disc estimates of minimum tenure requirement on pre-determined variables, majority MPs.

	(1) Age	(2) Female	(3) High school	(4) Degree	(5) South	(6) North	(7) Foreign	(8) Income (€)	(9) Private sector	(10) Public sector	(11) Self employed	(12) Out of labor force
Post*Tenure under 4.5 years	-0.409 (0.427)	-0.020 (0.017)	0.009 (0.019)	-0.001 (0.019)	0.019 (0.023)	-0.023 (0.023)	0.004 (0.006)	-12881.782 (8974.773)	0.040 (0.026)	-0.051** (0.023)	0.019 (0.015)	-0.008 (0.006)
N	22,785	22,785	15,140	15,140	22,785	22,785	22,785	22,728	14,290	14,290	14,290	14,290
R-squared	0.10	0.04	0.01	0.01	0.01	0.01	0.00	0.01	0.00	0.01	0.01	0.00
Average outcome	47.54	0.31	0.26	0.73	0.51	0.48	0.02	98716.98	0.55	0.33	0.10	0.02
MP FE	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
Party-by-gov. FE	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO

Notes: The regression sample is restricted to votes of confidence in a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. Age indicates the age when entering parliament for the first time (in years). No diploma, High school and University degree refer to the highest education level achieved. South, North and Foreign refer to the birthplace. Income refers to the private income in the year prior to entering Parliament and is available for MPs that enter Parliament later than 1981. Public sector, Private sector, Self employed and Out of the labor force refers to the MP occupation prior to entering Parliament. Data on education levels is not available for senators. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Table C5: Diff-in-disc estimates of minimum tenure requirement on pre-determined variables, opposition MPs.

	(1) Age	(2) Female	(3) High school	(4) Degree	(5) South	(6) North	(7) Foreign	(8) Income (€)	(9) Private sector	(10) Public sector	(11) Self employed	(12) Out of labor force
Post*Tenure under 4.5 years	-0.474 (0.594)	-0.003 (0.029)	0.014 (0.033)	-0.015 (0.036)	0.021 (0.036)	-0.018 (0.036)	-0.003 (0.009)	-8202.955 (5056.372)	-0.029 (0.041)	0.020 (0.040)	-0.010 (0.020)	0.019 (0.016)
N	9,221	9,221	5,970	5,970	9,221	9,221	9,221	9,193	5,492	5,492	5,492	5,492
R-squared	0.11	0.01	0.00	0.00	0.01	0.01	0.00	0.03	0.01	0.04	0.02	0.00
Average outcome	47.05	0.23	0.30	0.66	0.43	0.55	0.02	73767.88	0.50	0.36	0.12	0.02
MP FE	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
Party-by-gov. FE	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO

Notes: The regression sample is restricted to votes of confidence in a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. Age indicates the age when entering parliament for the first time (in years). No diploma, High school and University degree refer to the highest education level achieved. South, North and Foreign refer to the birthplace. Income refers to the private income in the year prior to entering Parliament and is available for MPs that enter Parliament later than 1981. Public sector, Private sector, Self employed and Out of the labor force refers to the MP occupation prior to entering Parliament. Data on education levels is not available for senators. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Table C6: Diff-in-disc estimates of minimum tenure requirement on confidence, by house.

	Chamber of Deputies			Senate		
	(1)	(2)	(3)	(4)	(5)	(6)
Post*Tenure under 4.5 years	0.083*** (0.021)	0.049*** (0.013)	0.031*** (0.007)	0.031 (0.035)	-0.014 (0.025)	0.037** (0.016)
N	22,635	22,635	22,635	10,403	10,403	10,403
R-squared	0.03	0.84	0.96	0.03	0.88	0.94
Average outcome	0.71	0.71	0.71	0.70	0.70	0.70
MP FE	NO	YES	YES	NO	YES	YES
Party-by-gov. FE	NO	NO	YES	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Table C7: Diff-in-disc estimates of minimum tenure requirement on confidence, local quadratic regressions.

	All			Majority party			Opposition party		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Post*Tenure under 4.5 years	-0.018 (0.027)	0.048*** (0.018)	0.039*** (0.007)	0.032** (0.014)	0.042*** (0.010)	0.035*** (0.009)	-0.040 (0.025)	0.041** (0.016)	0.023* (0.012)
N	33,038	33,038	33,038	22,785	22,785	22,785	9,221	9,221	9,221
R-squared	0.03	0.83	0.95	0.01	0.78	0.78	0.05	0.89	0.90
Average outcome	0.70	0.70	0.70	0.96	0.96	0.96	0.10	0.10	0.10
MP FE	NO	YES	YES	NO	YES	YES	NO	YES	YES
Party-by-gov. FE	NO	NO	YES	NO	NO	YES	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Table C8: Diff-in-disc estimates of minimum tenure requirement on abstension/no vote, all sample.

	(1)	(2)	(3)
Post*Tenure under 4.5 years	0.009 (0.019)	0.015 (0.018)	0.009 (0.019)
N	41,401	41,401	41,401
R-squared	0.01	0.30	0.31
Average outcome	0.20	0.20	0.20
MP FE	NO	YES	YES
Party-by-gov. FE	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Table C9: Diff-in-disc estimates of minimum tenure requirement on confidence, by type of party.

	Female			Male		
	(1)	(2)	(3)	(4)	(5)	(6)
Post*Tenure under 4.5 years	0.015	0.067	0.011	0.069***	0.061***	0.040***
	(0.061)	(0.043)	(0.010)	(0.021)	(0.015)	(0.009)
N	9,313	9,313	9,313	23,725	23,725	23,725
R-squared	0.05	0.86	0.96	0.02	0.82	0.95
Average outcome	0.75	0.75	0.75	0.69	0.69	0.69
MP FE	NO	YES	YES	NO	YES	YES
Party-by-gov. FE	NO	NO	YES	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Table C10: Diff-in-disc estimates of minimum tenure requirement on vote against party directives, by type of party.

	Female			Male		
	(1)	(2)	(3)	(4)	(5)	(6)
Post*Tenure under 4.5 years	-0.030**	-0.034**	-0.026	-0.047***	-0.036***	-0.029***
	(0.014)	(0.015)	(0.016)	(0.009)	(0.009)	(0.009)
N	11,373	11,373	11,373	28,515	28,515	28,515
R-squared	0.01	0.64	0.67	0.01	0.64	0.67
Average outcome	0.05	0.05	0.05	0.05	0.05	0.05
MP FE	NO	YES	YES	NO	YES	YES
Party-by-gov. FE	NO	NO	YES	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Table C11: Regression discontinuity estimates on confidence, heterogeneity by age of first year in parliament.

	Majority party		Opposition party	
	(1)	(2)	(3)	(4)
Tenure under 4.5 years	-0.057*	-0.030	0.119	0.113*
	(0.034)	(0.034)	(0.077)	(0.068)
Tenure under 4.5 years*age	0.002***	0.001**	-0.005***	-0.004***
	(0.001)	(0.001)	(0.002)	(0.001)
N	18,939	18,939	7,261	7,261
R-squared	0.01	0.06	0.02	0.19
MP FE	NO	NO	NO	NO
Party-by-gov. FE	NO	YES	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are Eicker-Huber-White heteroskedasticity-robust. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. Age is the age of MPs at the start of their parliamentary tenure (years). All regressions also control for a quadratic polynomial of age when entering parliament.

Table C12: Regression discontinuity estimates on confidence, heterogeneity by residualized pre-parliament income.

	Majority party		Opposition party	
	(1)	(2)	(3)	(4)
Tenure under 4.5 years	0.041***	0.041***	-0.091***	-0.084***
	(0.010)	(0.010)	(0.017)	(0.015)
Tenure under 4.5 years*income (million €)	-0.014	-0.013	0.012	-0.115
	(0.032)	(0.032)	(0.243)	(0.223)
N	18,886	18,886	7,222	7,222
R-squared	0.01	0.06	0.02	0.19
MP FE	NO	NO	NO	NO
Party-by-gov. FE	NO	YES	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. Income is the real income (in million $euro$ at 2005 prices) of the MP in the year before entering parliament, residualized by age. All regressions also control for a quadratic polynomial of age when entering parliament. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Table C13: Regression discontinuity estimates on votes against party directives, heterogeneity by age of first year in parliament.

	Majority party		Opposition party	
	(1)	(2)	(3)	(4)
Tenure under 4.5	0.030	0.011	0.137***	0.112**
years	(0.027)	(0.026)	(0.048)	(0.045)
Tenure under 4.5	-0.001**	-0.001	-0.004***	-0.003***
years*age	(0.001)	(0.001)	(0.001)	(0.001)
N	23,348	23,348	9,582	9,582
R-squared	0.00	0.04	0.02	0.12
MP FE	NO	NO	NO	NO
Party-by-gov. FE	NO	YES	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are Eicker-Huber-White heteroskedasticity-robust. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. Age is the age of MPs at the start of their parliamentary tenure (years). All regressions also control for a quadratic polynomial of age when entering parliament.

Table C14: Regression discontinuity estimates on votes against party directives, heterogeneity by residualized pre-parliament income.

	Majority party		Opposition party	
	(1)	(2)	(3)	(4)
Tenure under 4.5	-0.034***	-0.034***	-0.044***	-0.033***
years	(0.008)	(0.008)	(0.012)	(0.011)
Tenure under 4.5	0.012	0.011	-0.046	-0.041
years*income (million €)	(0.025)	(0.026)	(0.156)	(0.161)
N	23,275	23,275	9,524	9,524
R-squared	0.00	0.04	0.01	0.12
MP FE	NO	NO	NO	NO
Party-by-gov. FE	NO	YES	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. Income is the real income (in million $euro$ at 2005 prices) of the MP in the year before entering parliament, residualized by age. All regressions also control for a quadratic polynomial of age when entering parliament. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Table C15: Regression discontinuity estimates on votes against party directives, heterogeneity by majority margin.

	Majority party			Opposition party		
	(1)	(2)	(3)	(4)	(5)	(6)
Tenure under 4.5 years	-0.072***	-0.066***	-0.065***	-0.060***	-0.006	0.004
	(0.018)	(0.017)	(0.017)	(0.021)	(0.018)	(0.017)
Tenure under 4.5 years*Majority margin ('00)	0.025***	0.023***	0.023***	0.031**	-0.016	-0.008
	(0.008)	(0.008)	(0.008)	(0.012)	(0.010)	(0.008)
N	23,348	23,348	23,348	9,566	9,566	9,566
R-squared	0.01	0.55	0.55	0.02	0.78	0.79
MP FE	NO	YES	YES	NO	YES	YES
Party FE	NO	NO	YES	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. Majority margin indicates the majority margin in the house in which the confidence vote took place measured in hundred MPs. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Table C16: Diff-in-disc estimates of minimum tenure requirement on party-switches by party of origin and destination, all sample.

	From majority to majority			From opposition to majority		
	(1)	(2)	(3)	(4)	(5)	(6)
Post*Tenure under 4.5 years	0.00023***	0.00021***	0.00021***	0.00001***	0.00002***	0.00002***
	(0.00003)	(0.00003)	(0.00003)	(0.00000)	(0.00001)	(0.00001)
N	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204
R-squared	0.00009	0.00216	0.00219	0.00001	0.00182	0.00188
Average outcome	0.00006	0.00006	0.00006	0.00001	0.00001	0.00001
MP FE	NO	YES	YES	NO	YES	YES
Party-by-gov. FE	NO	NO	YES	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Table C17: Diff-in-disc estimates of minimum tenure requirement on party-switches by party of origin and destination, all sample.

	From majority to opposition			From opposition to opposition		
	(1)	(2)	(3)	(4)	(5)	(6)
Post*Tenure under 4.5 years	0.00002	0.00002*	0.00003*	-0.00001	-0.00001	0.00000
	(0.00001)	(0.00001)	(0.00001)	(0.00002)	(0.00002)	(0.00002)
N	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204
R-squared	0.00001	0.00185	0.00216	0.00003	0.00172	0.00177
Average outcome	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001
MP FE	NO	YES	YES	NO	YES	YES
Party-by-gov. FE	NO	NO	YES	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

Table C18: Diff-in-disc estimates of minimum tenure requirement on *ordinary* law proposed and approved.

	Ordinary law proposed			Ordinary law proposed and approved		
	(1)	(2)	(3)	(4)	(5)	(6)
Post*Tenure under 4.5 years	-0.00051	0.00028	0.00022	0.00011	0.00036***	0.00047***
	(0.00046)	(0.00047)	(0.00053)	(0.00012)	(0.00013)	(0.00014)
N	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204
R-squared	0.001	0.009	0.010	0.000	0.005	0.006
Average outcome	0.0036	0.0036	0.0036	0.0003	0.0003	0.0003
MP FE	NO	YES	YES	NO	YES	YES
Party-by-gov. FE	NO	NO	YES	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

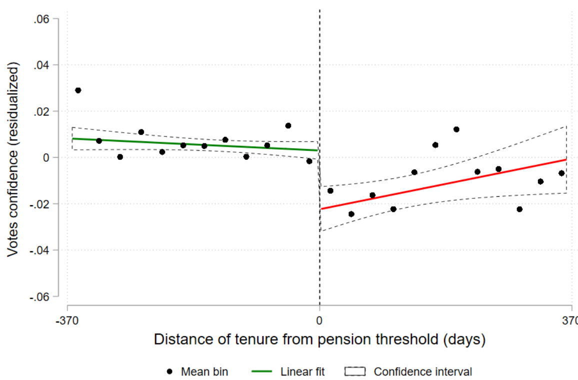
Table C19: Diff-in-disc estimates of minimum tenure requirement on *constitutional* law proposed and approved.

	Constitutional law proposed			Constitutional law proposed and approved		
	(1)	(2)	(3)	(4)	(5)	(6)
Post*Tenure under 4.5 years	0.00022***	0.00025***	0.00032***	0.00005**	0.00005**	0.00006**
	(0.00008)	(0.00008)	(0.00010)	(0.00002)	(0.00003)	(0.00003)
N	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204	1,290,204
R-squared	0.000	0.004	0.005	0.000	0.004	0.004
Average outcome	0.0002	0.0002	0.0002	0.0000	0.0000	0.0000
MP FE	NO	YES	YES	NO	YES	YES
Party-by-gov. FE	NO	NO	YES	NO	NO	YES

Notes: The regression sample is restricted to a bandwidth of 12 months on each side of the cutoff. Standard errors are clustered at MP level. Average outcome is the average of the outcome variable within the bandwidth after reaching the tenure threshold. In all regressions we control for the house of parliament (Chamber or Senate) of the MP.

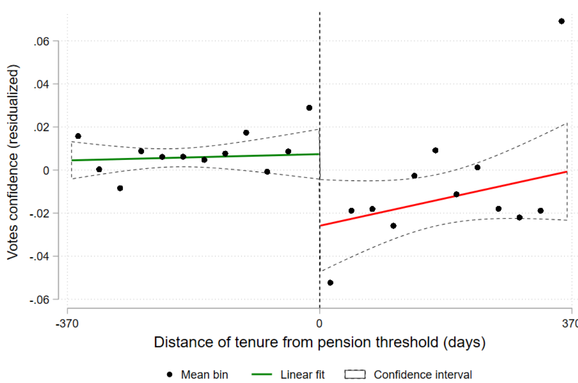
Appendix G Additional Figures

Figure D3: Regression discontinuities by House, post-treatment and pre-treatment



(a) Chamber of Deputies, post-treatment

(b) Chamber of Deputies, pre-treatment

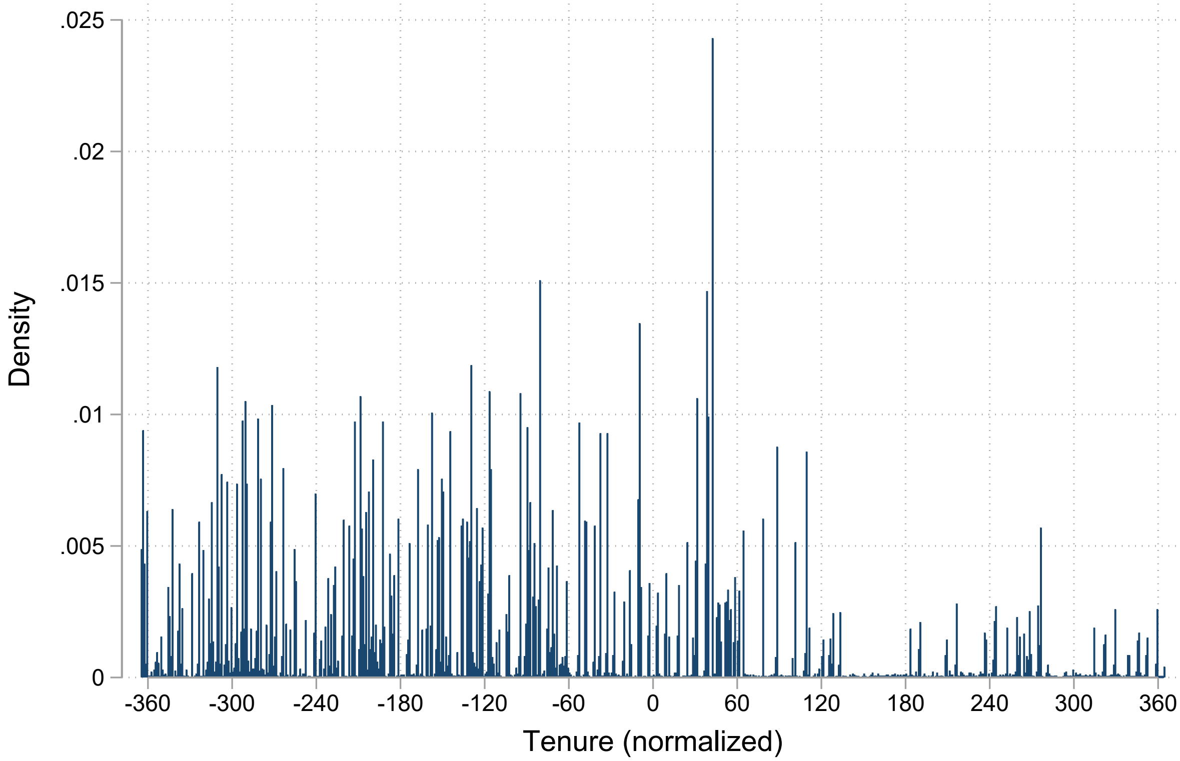


(c) Senate, post-treatment

(d) Senate, pre-treatment

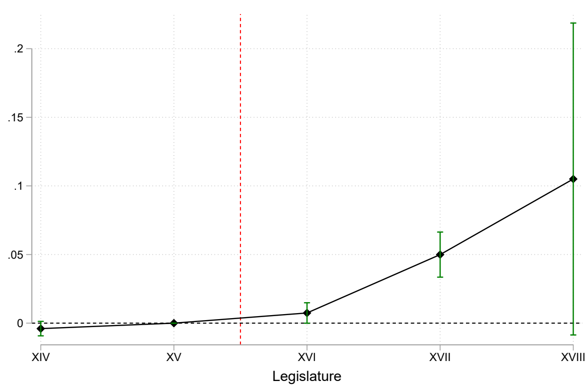
Notes: These figures show the effect of the parliamentary tenure distance from the 4.5 year-cutoff on the MP's probability of voting confidence in the government. The circles are averages across monthly bins on either side of the threshold, while the solid and dashed lines represent the predicted values and confidence intervals of a local linear regression of the outcome on (days of tenure, normalized) and the fixed effects, separately for each side of the cutoff. The bandwidth includes observations within one year from the 4.5 year-cutoff.

Figure D4: Density of confidence votes

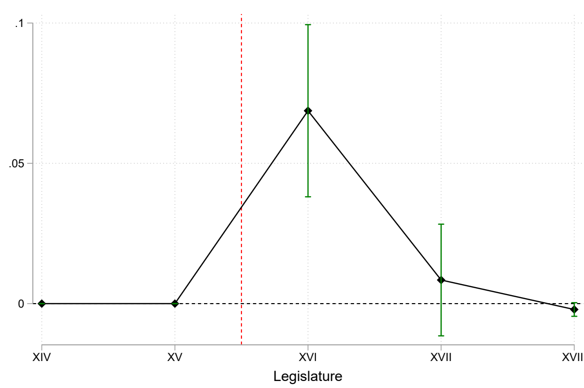


Notes: This figure plots the density of confidence votes in daily bins for both Houses over the number of days of tenure from the 4.5-year threshold.

Figure D5: RD coefficients, by legislature and House



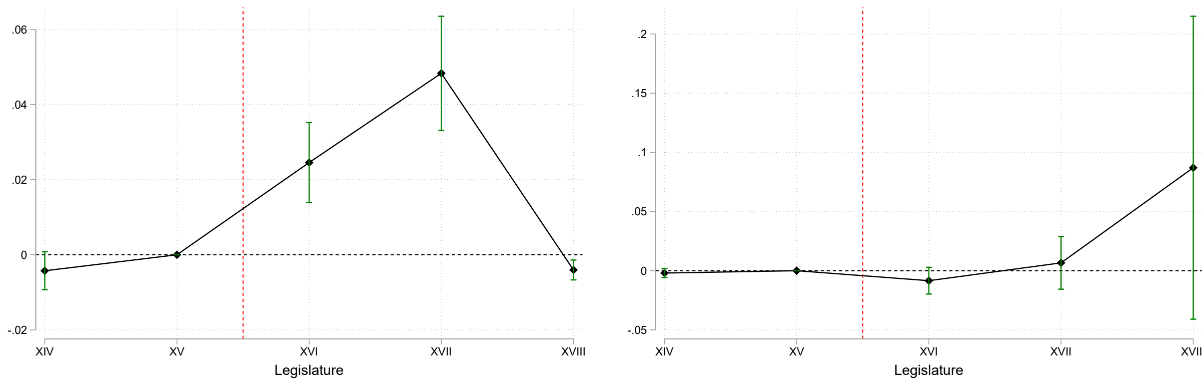
(a) Chamber of Deputies



(b) Senate

Notes: These figures show the RD coefficient and its 95% confidence interval estimating regression Equation (5), separately for each House and legislature (XIV-XVIII).

Figure D6: RD coefficients, by party's stance with respect to the current government

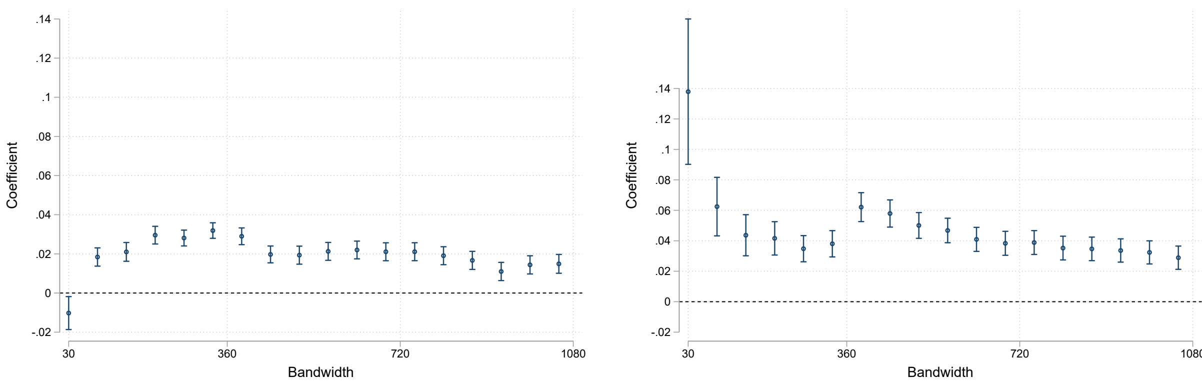


(a) Elected in majority party

(b) Elected in opposition party

Notes: These figures show the RD coefficient and its 95% confidence interval estimating regression Equation (5), separately for MPs elected in parties supporting the government and MPs elected in parties opposing the government.

Figure D7: Diff-in-disc coefficients: bandwidth sensitivity, by House

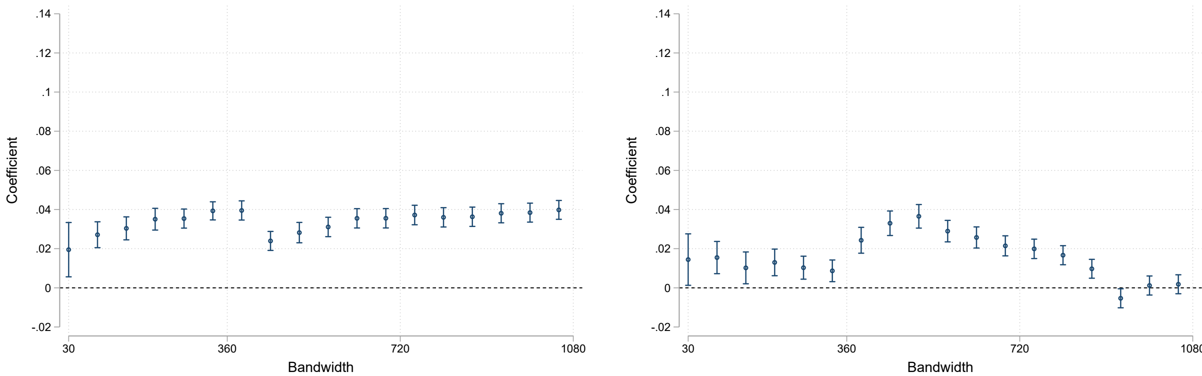


(a) Chamber of Deputies

(b) Senate

Notes: The graph shows the point estimate and the 95% confidence interval of the Diff-in-Disc coefficients on the entire sample (separately for each House), estimating the regression specified in Equation (6) using different bandwidths. The horizontal axis shows the number of tenure days from the 4.5-year-cutoff in each side of the bandwidth.

Figure D8: Diff-in-disc coefficients: bandwidth sensitivity, by party's stance with respect to the current government.

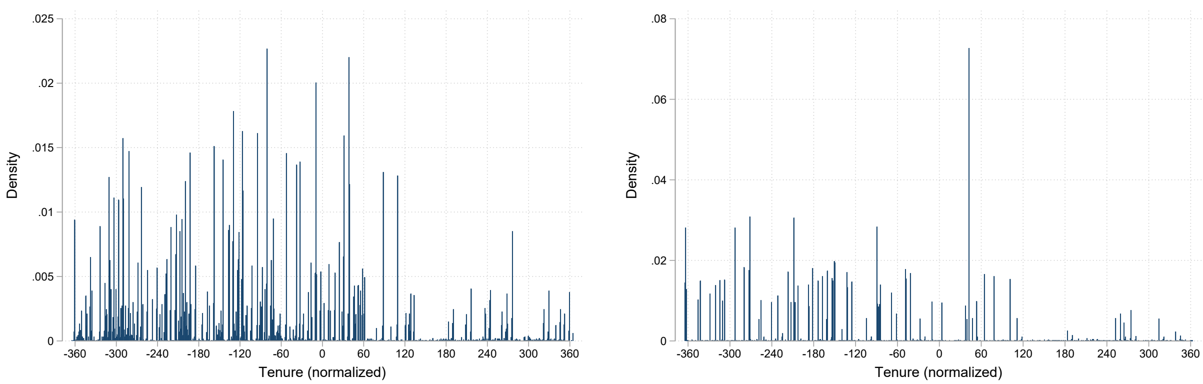


(a) Elected in majority party

(b) Elected in opposition party

Notes: The graph shows the point estimate and the 95% confidence interval of the Diff-in-Disc coefficients on the entire sample (separately for majority and opposition MPs), estimating the regression specified in Equation (6) using different bandwidths. The horizontal axis shows the number of tenure days from the 4.5-year-cutoff in each side of the bandwidth.

Figure D9: Density of confidence votes, by House

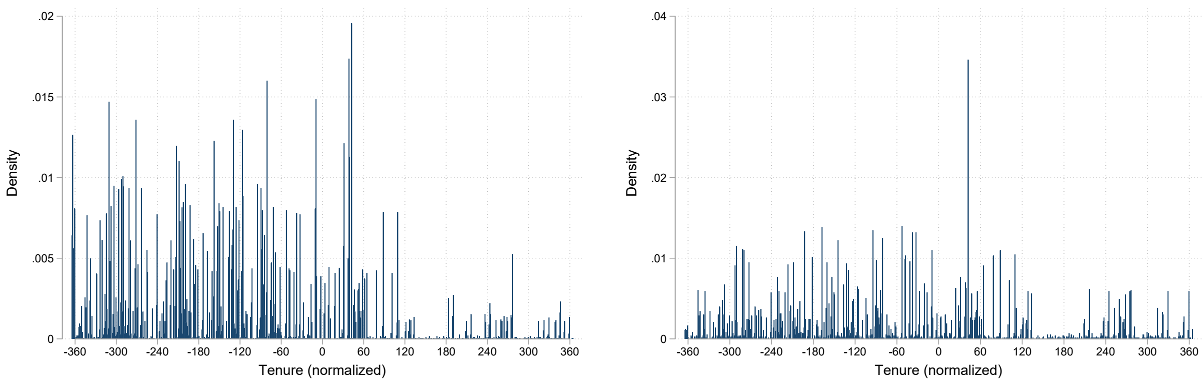


(a) Chamber of Deputies

(b) Senate

Notes: This figure plots the density of confidence votes in daily bins over the number of days of tenure from the 4.5-year threshold, separately for each House.

Figure D10: Density of confidence votes, by party's stnace with respect to the current government.

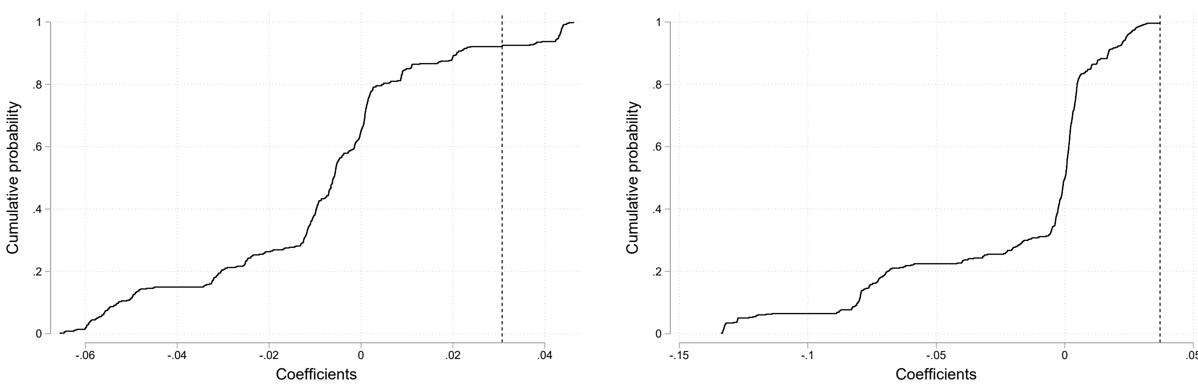


(a) Elected in majority party

(b) Elected in opposition party

Notes: This figure plots the density of confidence votes in daily bins over the number of days of tenure from the 4.5-year threshold, separately for majority and opposition MPs.

Figure D11: Placebo estimates, by House

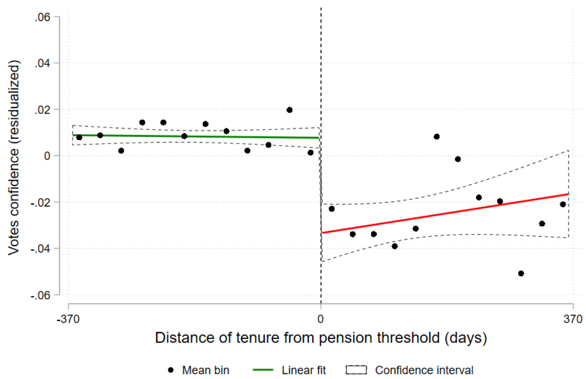


(a) Chamber of Deputies

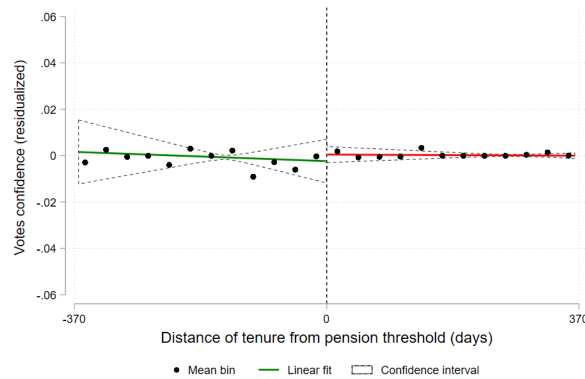
(b) Senate

Notes: This graph shows the cumulative distribution of Diff-in-Disc estimates on votes of confidence for each House separately, from placebo local linear regressions in which the cutoff is set in different parts of the tenure distribution. Estimates are computed using the regression in Equation (6) within a 1-year bandwidth. Cut-offs are located at every day from 3 years and 7 months to 4 years and 5 months and at every day from 4 years and 7 months to 5 years and 5 months, in order to stay sufficiently away from the policy thresholds of 3.5, 4.5 and 5.5 years at which the severance payment, the pension age and the pension payment change discontinuously. The vertical dashed line shows the coefficient estimated using the true 4.5-year tenure threshold.

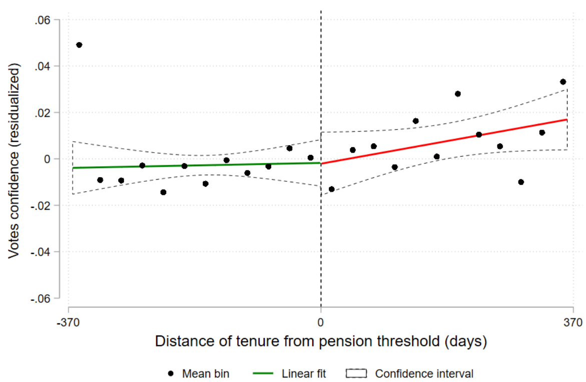
Figure D12: Regression discontinuities, by party's stance with respect to the current government.



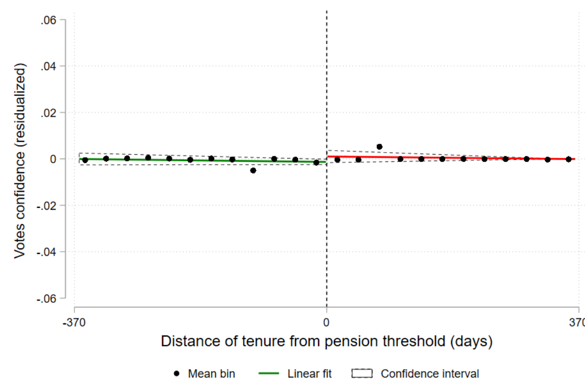
(a) Elected in majority party, post-treatment



(b) Elected in majority party, pre-treatment



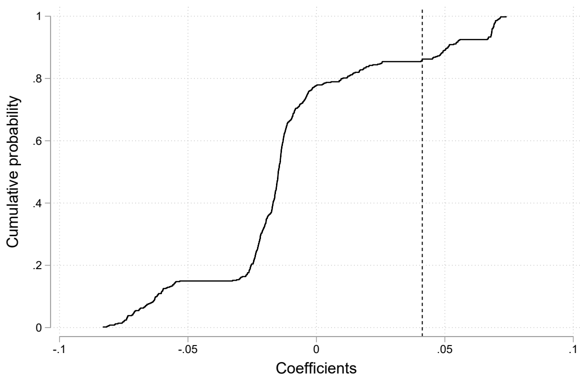
(c) Elected in opposition party, post-treatment



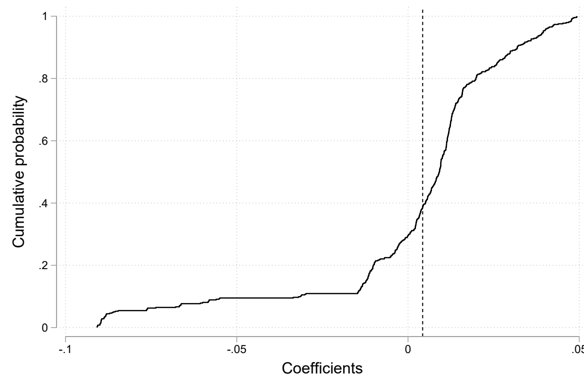
(d) Elected in opposition party, pre-treatment

Notes: These figures show the effect of the parliamentary tenure distance from the 4.5 year-cutoff on the MP's probability of voting confidence in the government. The circles are averages across monthly bins on either side of the threshold, while the solid and dashed lines represent the predicted values and confidence intervals of a local linear regression of the outcome on (days of tenure, normalized) and the fixed effects, separately for each side of the cutoff. The bandwidth includes observations within one year from the 4.5 year-cutoff.

Figure D13: Placebo estimates, by party's stance with respect to the current government.



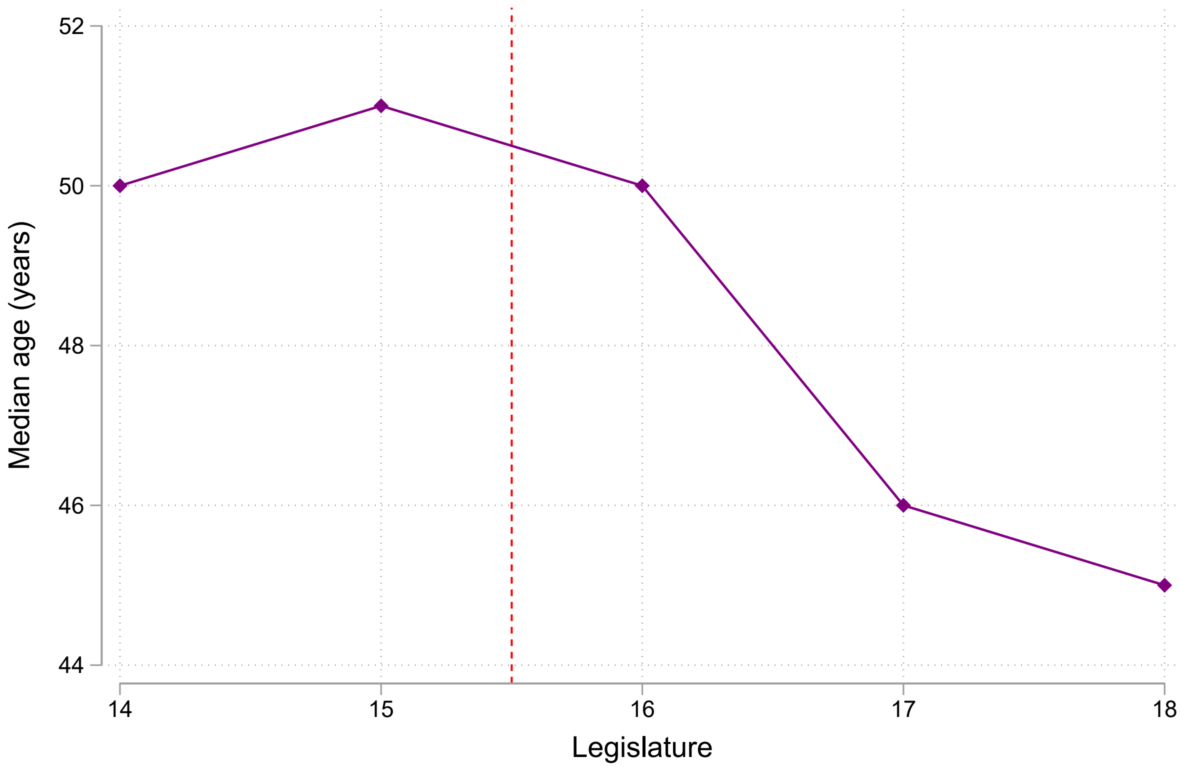
(a) Elected in majority party



(b) Elected in opposition party

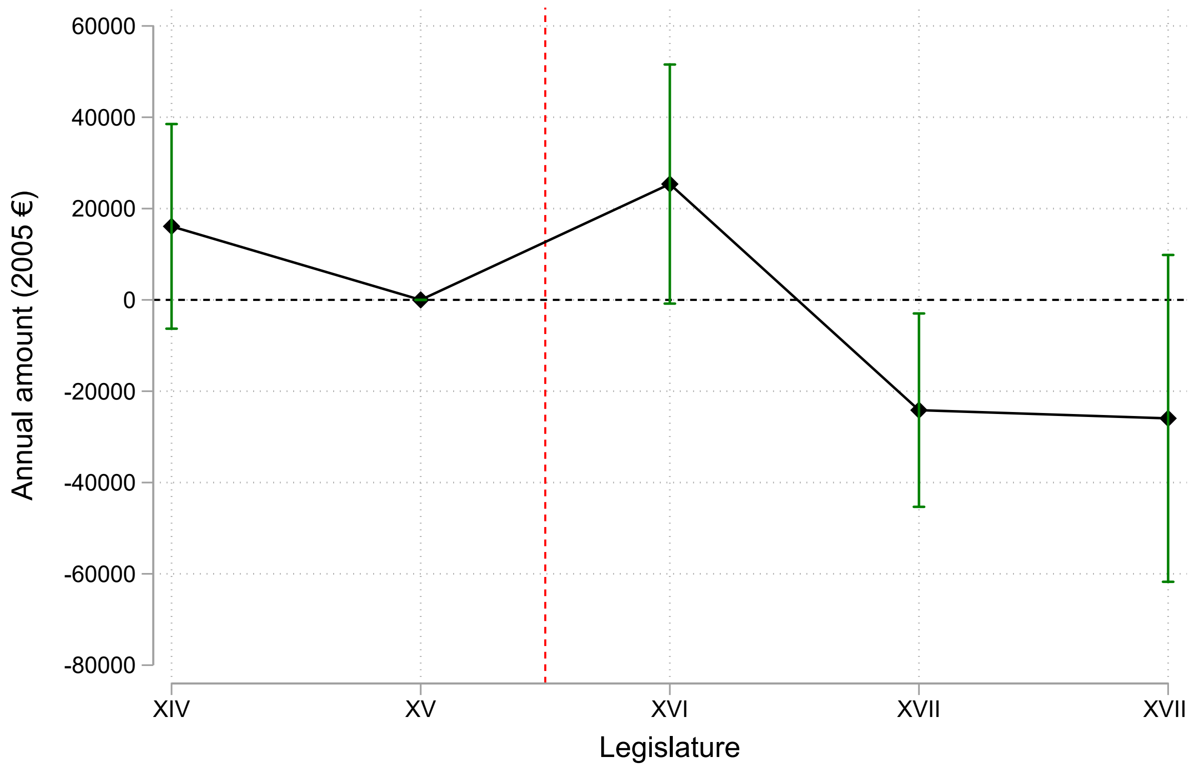
Notes: This graph shows the cumulative distribution of Diff-in-Disc estimates on votes of confidence for majority and opposition MPs separately, from placebo local linear regressions in which the cutoff is set in different parts of the tenure distribution. Estimates are computed using the regression in Equation (6) within a 1-year bandwidth. Cut-offs are located at every day from 3 years and 7 months to 4 years and 5 months and at every day from 4 years and 7 months to 5 years and 5 months, in order to stay sufficiently away from the policy thresholds of 3.5, 4.5 and 5.5 years at which the severance payment, the pension age and the pension payment change discontinuously. The vertical dashed line shows the coefficient estimated using the true 4.5-year tenure threshold.

Figure D14: Median age, by legislature (both Houses)



Notes: This figure shows the median age when entering parliament for newly-elected MPs in each legislature.

Figure D15: Pre-parliament income variation across legislatures controlling for age



Notes: This figure shows the estimates for the coefficients λ_j and their 95 percent confidence interval in the regression $w_{ij} = \alpha_0 + \alpha_1 age_{ij} + \alpha_2 age^2_{ij} + \sum_{j \neq XV} \lambda_j legislature_j + \nu_{ij}$.
w_{ij} is the pre-parliament wage of MP i elected for the first time in legislature j. age_{ij} is the age of MP i elected for the first time in legislature j when entering parliament. $legislature_j$ is a fixed effect for legislature j. ν_{ij} is the error term.

Figure D16: Cdfs of annual income in the year before becoming MP, by legislature

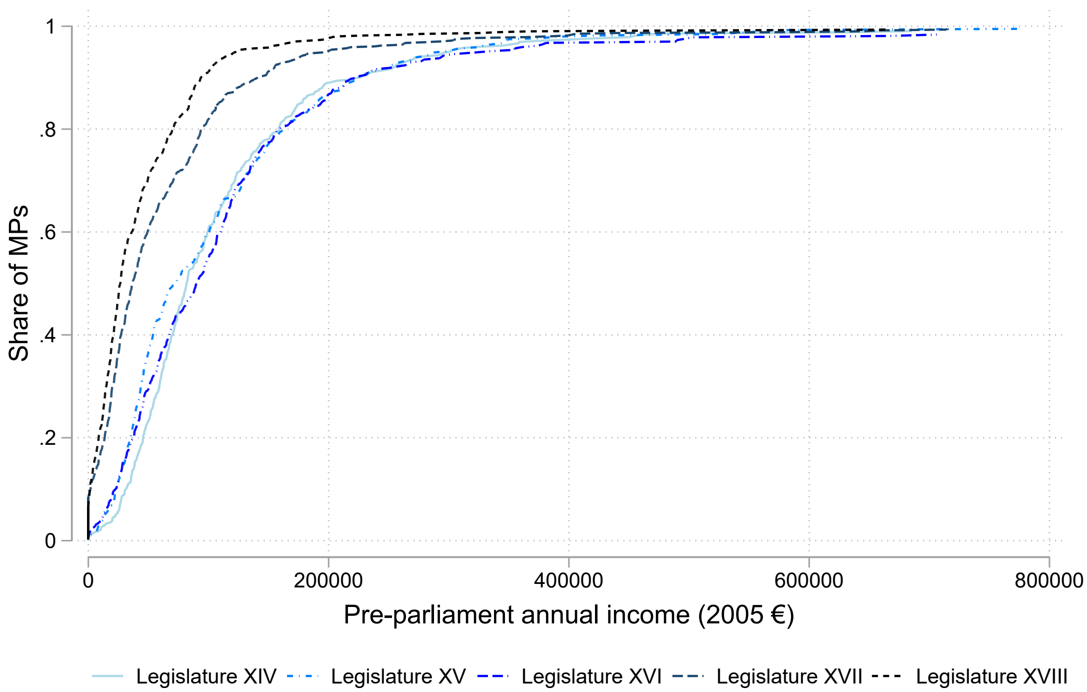
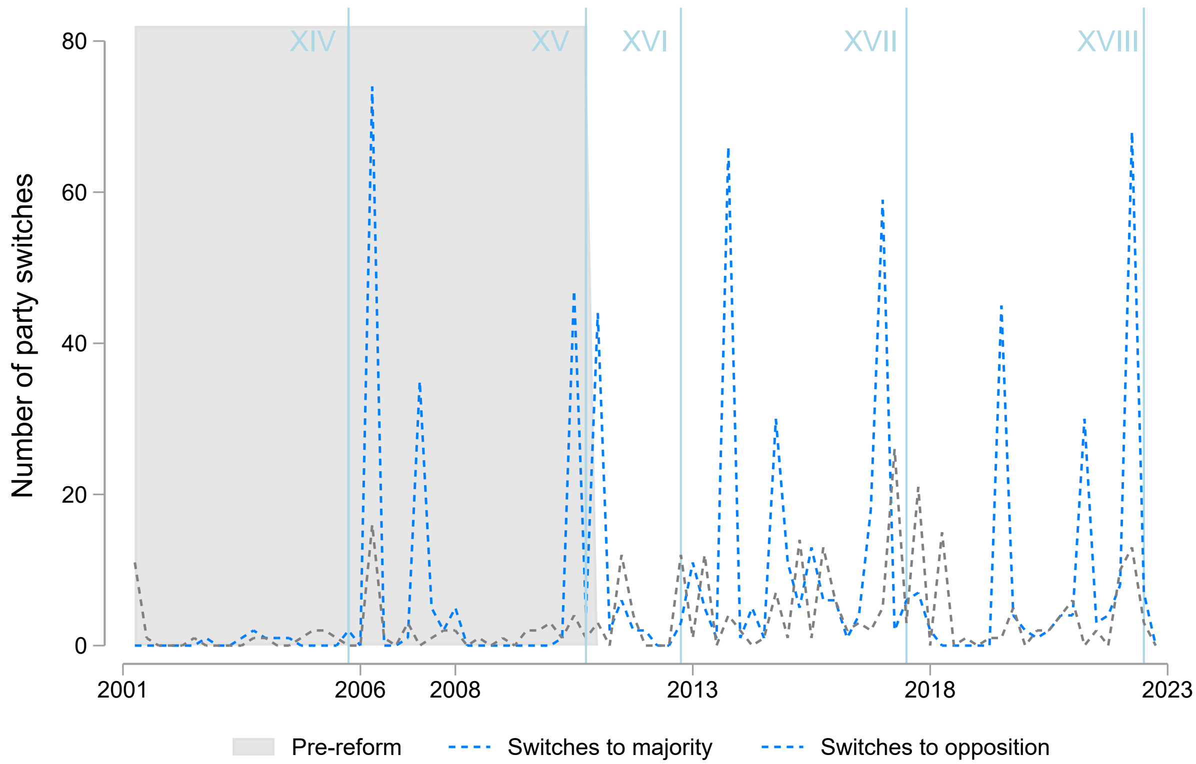


Figure D17: Party switches over time



Notes: The graph shows the number of MPs' switches to majority parties (blue) and opposition parties (gray) in each quarter of legislature XIV-XVIII. The light blue lines are placed 4.5 years after the beginning of each legislature. The pre-reform period (gray area) represents the period up to 4.5 years after the beginning of last legislature before the introduction of the minimum tenure requirement (XV).